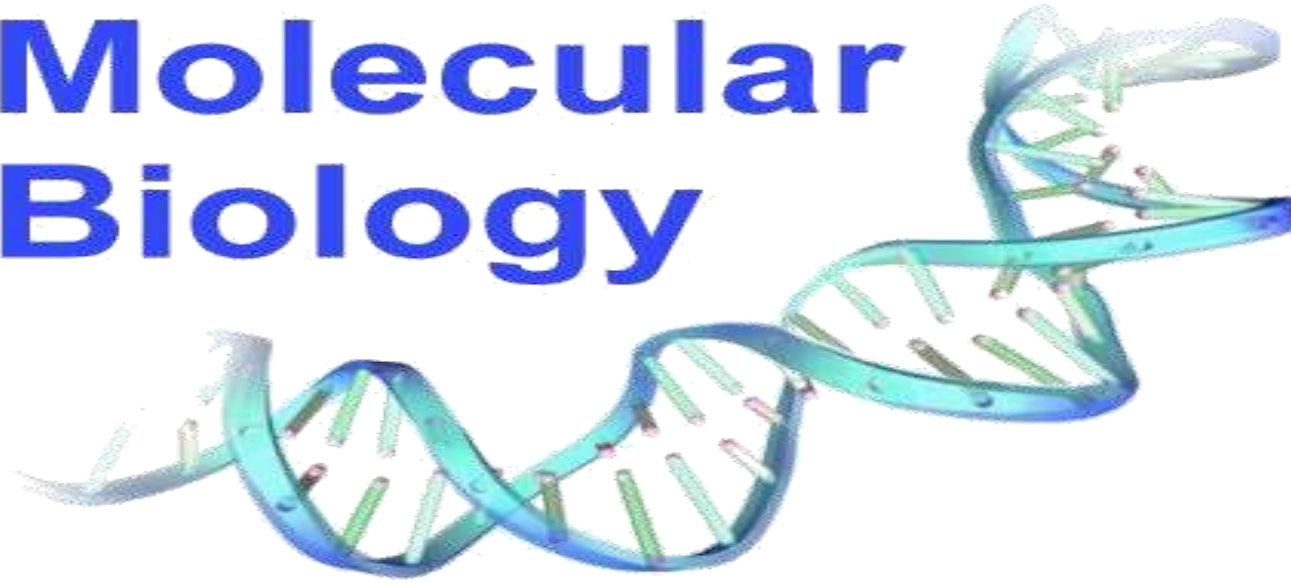


Chapter 2

Molecular Biology



Tamrat Nida (M.Sc)
Dep't of Med. Biochemistry,
AAU, CHS, SM

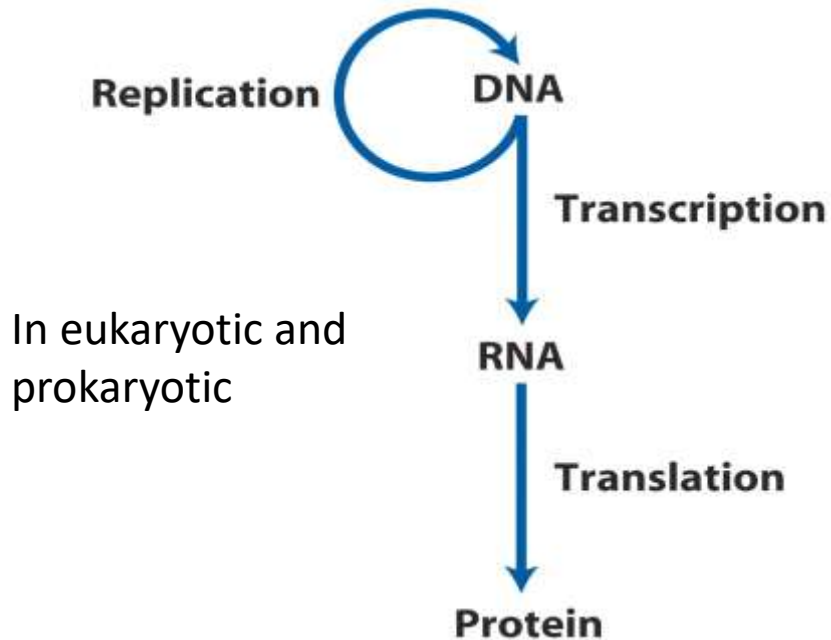
Outline

- Nucleotide structure
- Biosynthesis & Degradation of nucleotides
- DNA structure and Replication
- RNA structures and Transcription
- Protein Synthesis:
 - The Genetic Code
 - Translation
 - Mutation
- Regulation of gene expression in
 - Prokaryotes
 - Eukaryotes

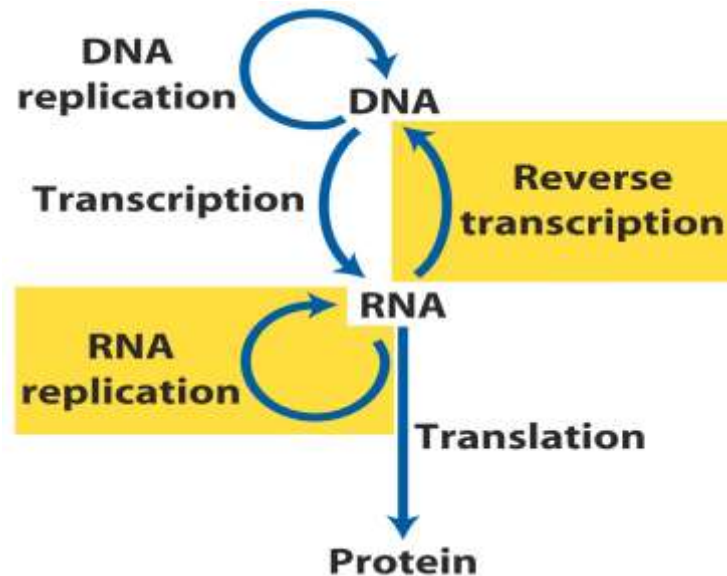
Introduction

- The genetic information is stored in **DNA** or **RNA** (RNA viruses)
- **The central dogma of biochemistry**; is the **flow of genetic information** from **DNA** to DNA from DNA to **RNA**, and then to **protein**

Traditional dogma



Extended dogma



- **RNA viruses like HIV** contain their **genetic information as RNA**, The info is reversely transcribed into DNA (by **reverse transcriptase enzyme**) so that it can be integrated with the host tissue

Signíficances of nucleotídes

- They are **precursors** of DNA and RNA
- They are the **energy currency** in metabolic transactions, as a triphosphate (ATP and GTP) has highly energetic phosphate
- They are **components of:**
 - **Cofactors:** such as NAD, FAD, S-adenosylmethionine, and coenzyme A.
 - **Activated biosynthetic intermediates:** such as UDP-glucose and CDP-diacylglycerol
 - **Second messengers:** such as cAMP and cGMP
 - i.e. Mediate many **hormonal signals**.

Role of Nucleotides

Partially **regulate** metabolic pathways:-

Messenger RNA (mRNA) directs ribosomal protein synthesis

Transfer RNA (tRNA) transports amino acids in the cell and delivers them to the ribosome

Ribosomal RNA (rRNA) makes up about 2/3 of the ribosome

structural and functional roles:-

Ribozymes have catalytic properties – function like an enzyme

DNA stores the genetic code

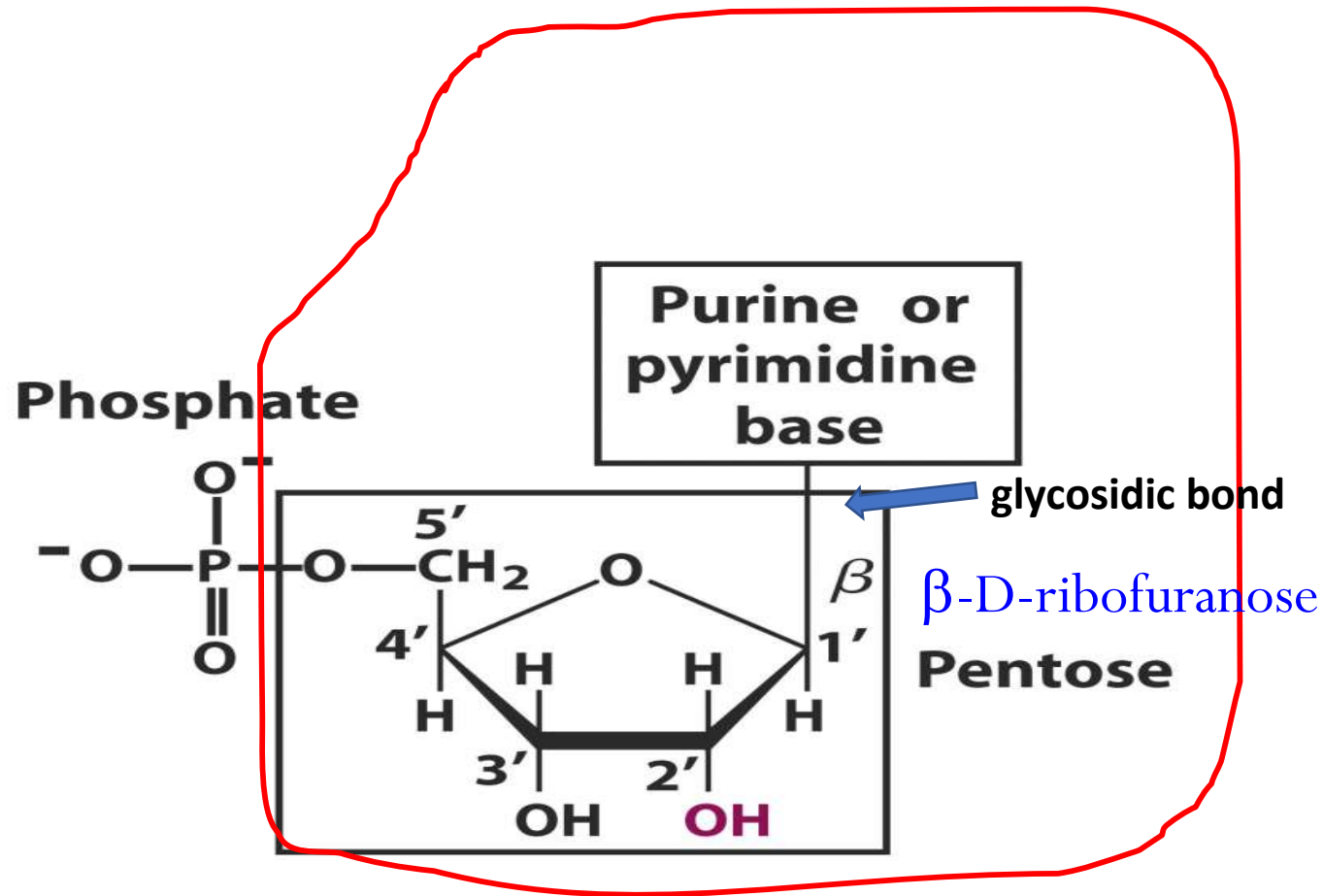
TERMINOLOGY OF NUCLEIC ACIDS

Bases	Purine (Adenine [A], Guanine [G]) Pyrimidine (Uracil [U], Cytosine [C], Thymine [T])
Sugar	Deoxyribose (DNA), Ribose (RNA)
Base + Sugar	Nucleoside
Base + Sugar + phosphate	Nucleotide

Structures of Nucleotides- (D(R)NA

Building Blocks

Without the phosphoryl group, this would be called a *nucleoside*



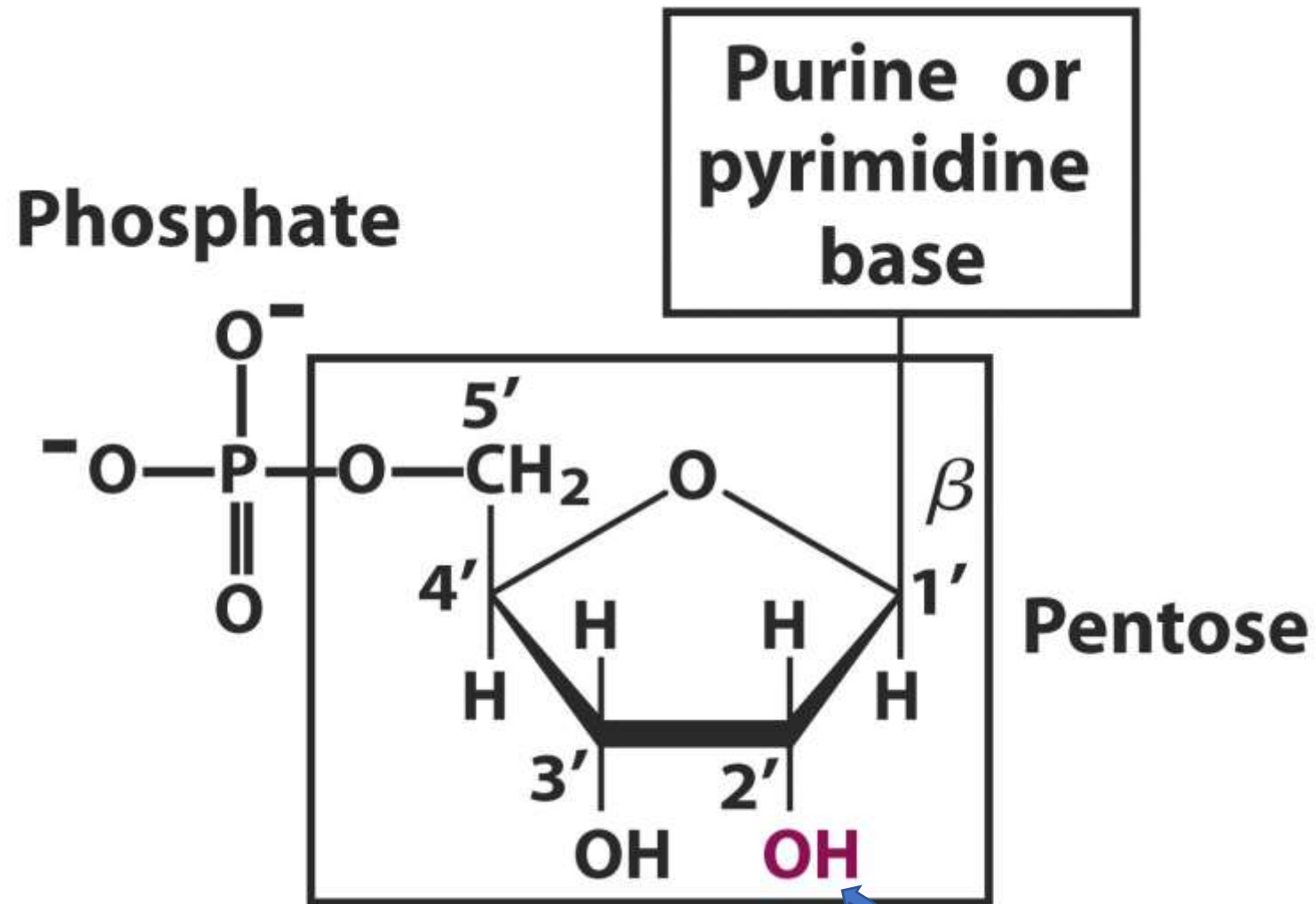
Nucleotides- → Are the basic building block of nucleic acid (RNA and DNA)

Structurally have 3 components

Nitrogenous bases + Pentose = Nucleosides

Nucleoside + Pi = Nuclotide

Nucleotide



In RNA, the sugar is ribose (OH)

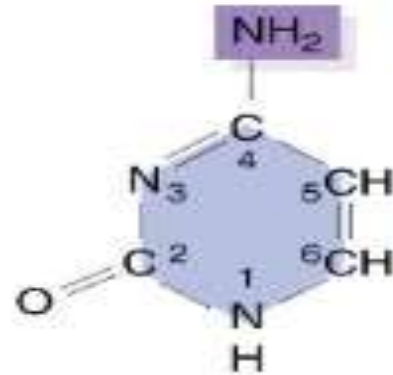
In DNA, the sugar is deoxyribose (H).

Nitrogenous bases

Nitrogenous bases:- they are derived from **2 parent compounds.. Purine (double)** and **pyrimidine ring (single)**



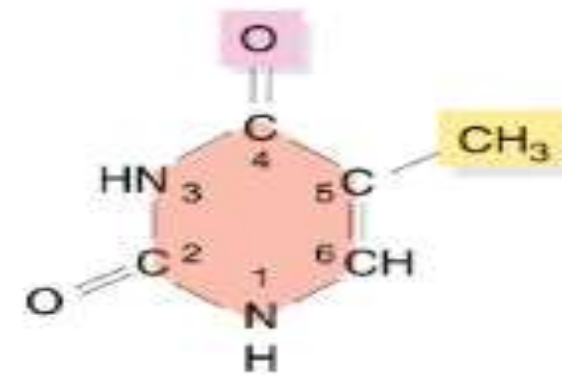
Pyrimidine



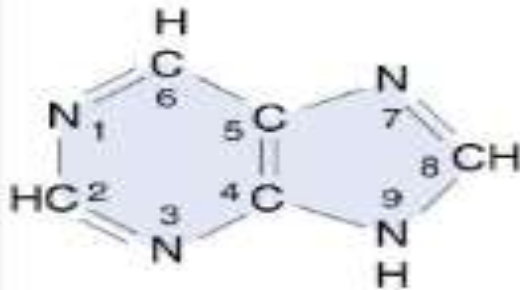
Cytosine (C)



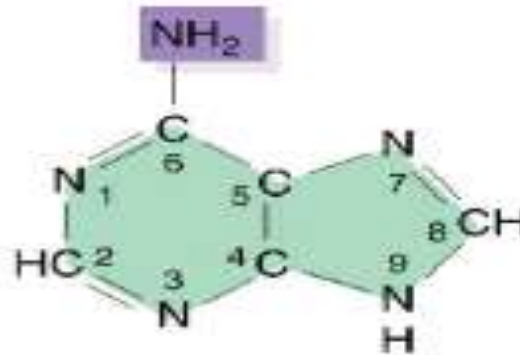
Uracil (U)
(found in RNA)



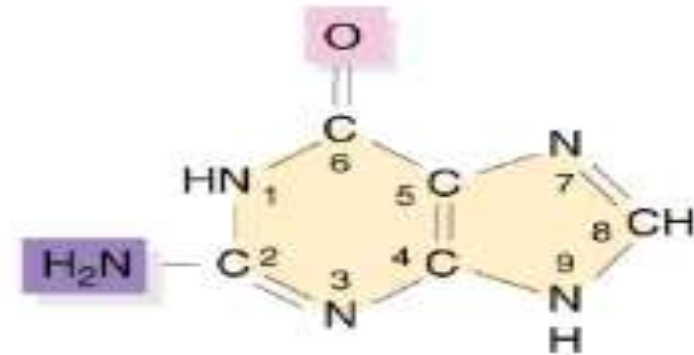
Thymine (T)
(found in DNA)



Purine



Adenine (A)



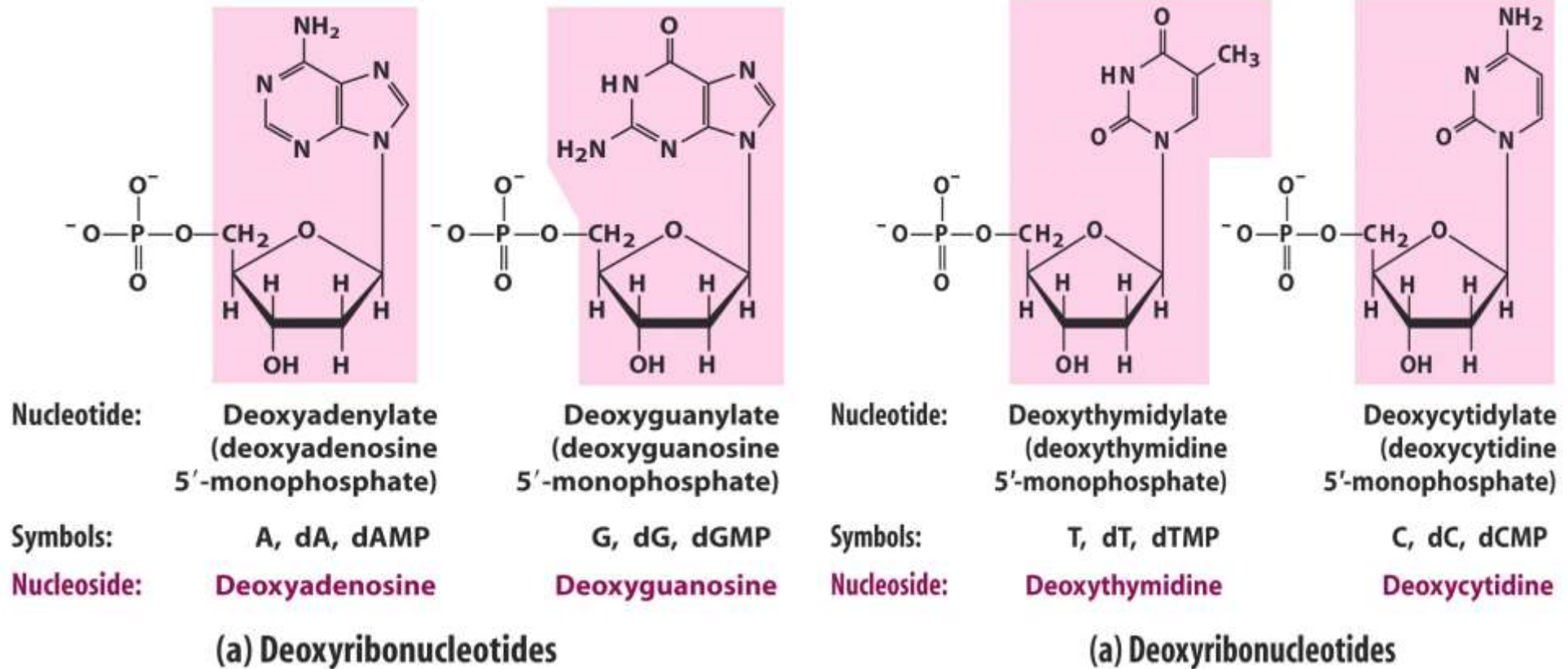
Guanine (G)

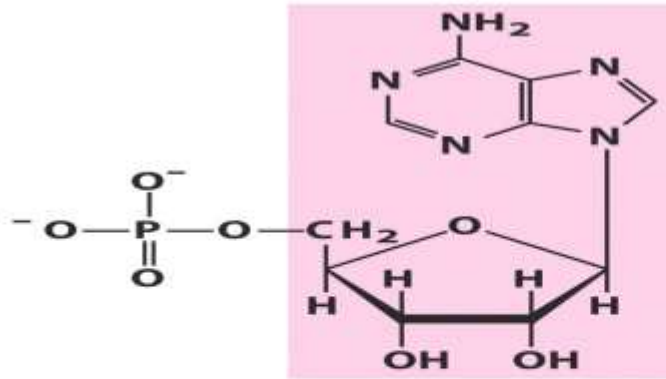
Nomenclature

A nucleoside is shown in pink

The **nucleotide** is the **entire structure with at least one phosphate**

“Nucleotide = **Nucleoside** monophosphate”





Nucleotide:

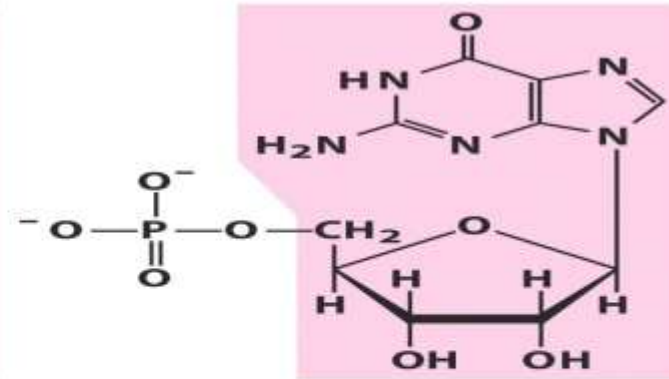
Adenylate (adenosine 5'-monophosphate)

Symbols:

A, AMP

Nucleoside:

Adenosine

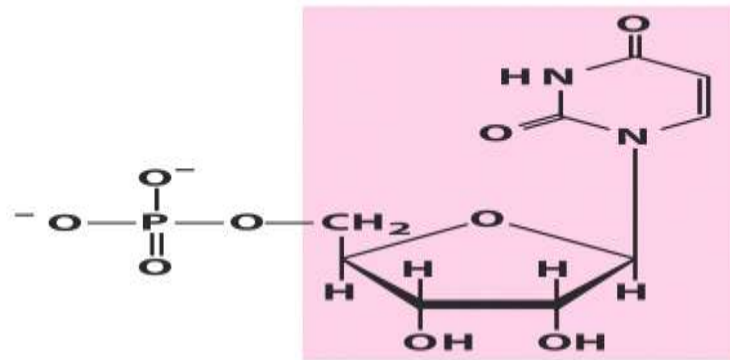


Guanylate (guanosine 5'-monophosphate)

G, GMP

Guanosine

(b) Ribonucleotides



Nucleotide:

Uridylate (uridine 5'-monophosphate)

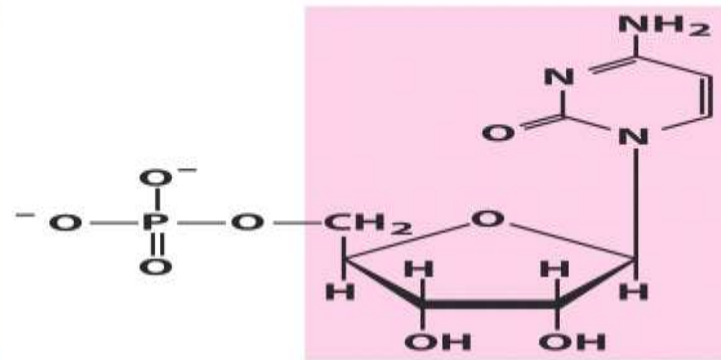
Symbols:

U, UMP

Nucleoside:

Uridine

(b) Ribonucleotides



Cytidylate (cytidine 5'-monophosphate)

C, CMP

Cytidine

“Initially, they were
Synthesis as Ribo
Nucleotide .. later
converted to deoxy
nucleotide by a
“nucleotide reductase
enzyme” removal of the
2' OH

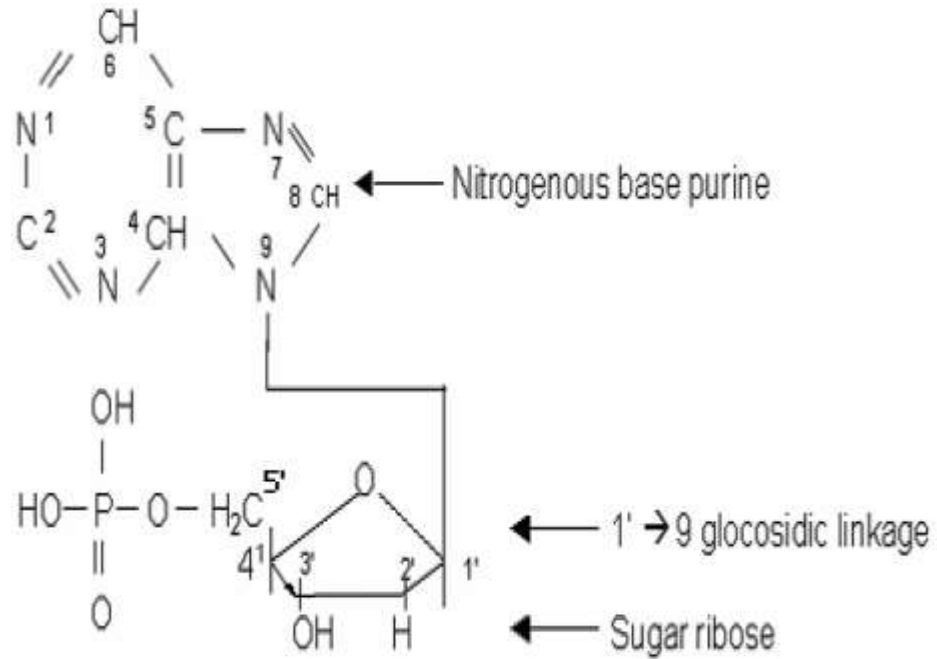


Fig . Purine deoxyribonucleotide

Purine double-membered has a **higher number of atoms** in their ring and **requires higher energy (ATP)** to **synthesize** them (11 steps)

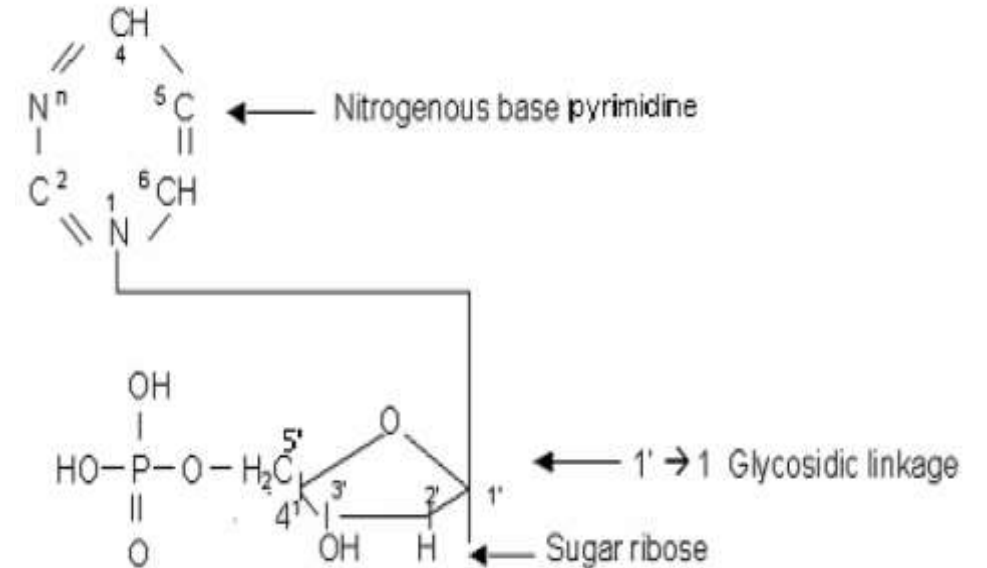


Fig. Pyrimidine deoxyribonucleotide

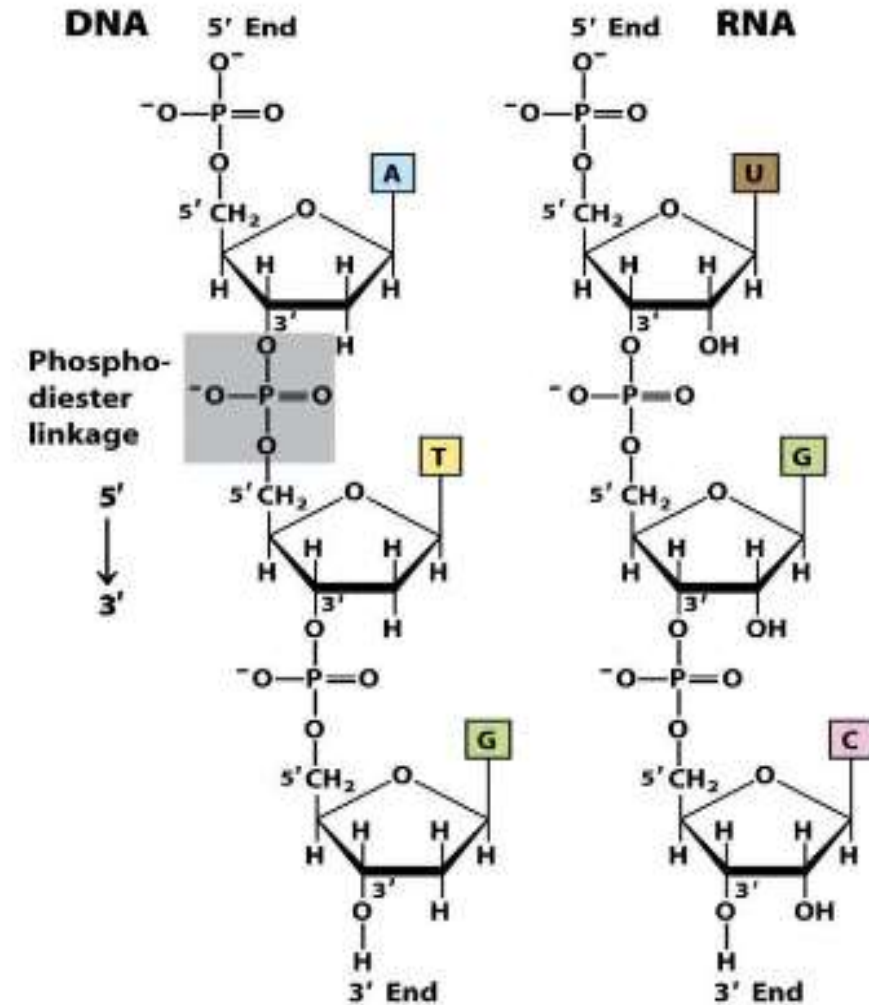
Pyrimidines are **single-membered** (6-step reaction) (C1 to N1 glycosidic Bond)

Nucleotide and Nucleic Acid Nomenclature

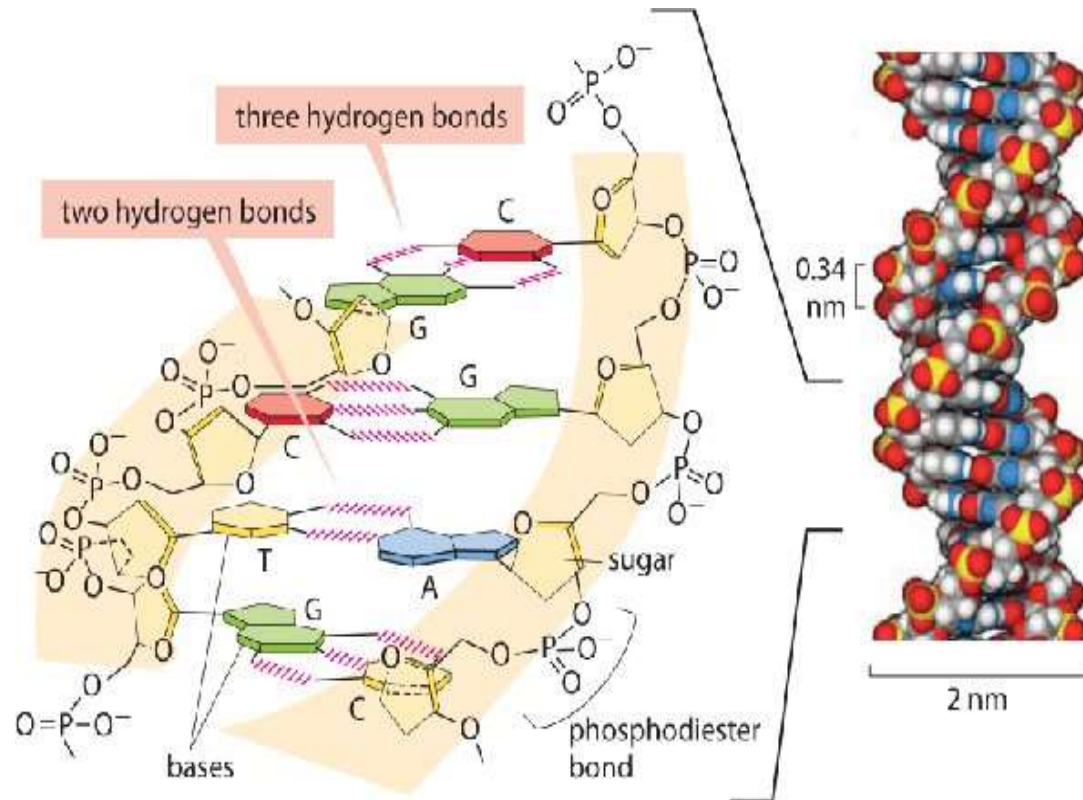
<i>Base</i>	<i>Nucleoside</i>	<i>Nucleotide</i>	<i>Nucleic acid</i>
Purines			
Adenine	Adenosine	Adenylate	RNA
	Deoxyadenosine	Deoxyadenylate	DNA
Guanine	Guanosine	Guanylate	RNA
	Deoxyguanosine	Deoxyguanylate	DNA
Pyrimidines			
Cytosine	Cytidine	Cytidylate	RNA
	Deoxycytidine	Deoxycytidylate	DNA
Thymine	Thymidine or deoxythymidine	Thymidylate or deoxythymidylate	DNA
Uracil	Uridine	Uridylate	RNA

Phosphodiester bond

- Links successive nucleotide in nucleic acids(DNA and RNA)
- Covalently linked through phosphate- group bridge
- 5' phosphate group of one nucleotide links with the 3' hydroxyl group of the next nucleotide



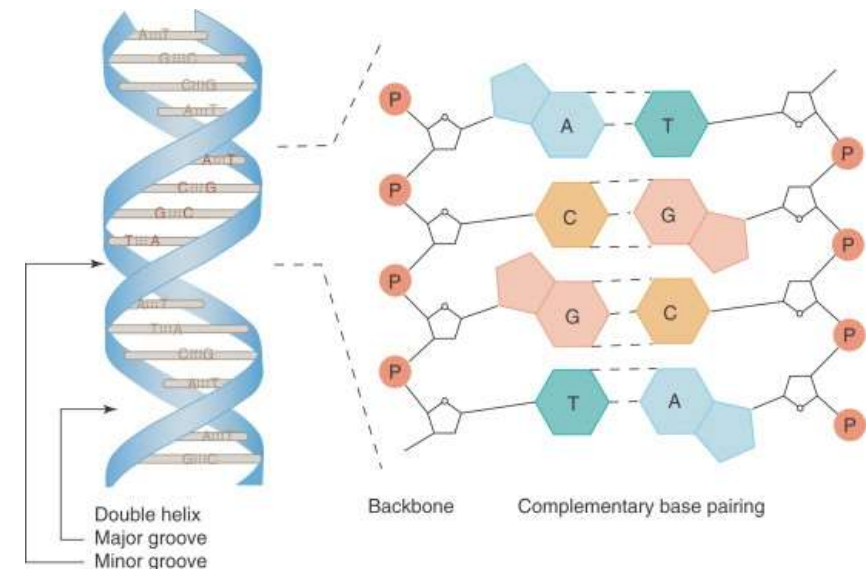
H- Bond-associated complementarity



- The **backbone (Sugar and phosphate)** of both DNA and RNA is **hydrophilic** it is **outward**
- **Base pair is inward (fear water)** in the middle The base
- Purine and pyrimidine are **hydrophobic (insoluble in water)** at near neutral Ph of the cell (but in **acidic** and **basic Ph** bases become **charged** and their **solubility increases**)



Watson and crick (1953) → double helix



The major groove is the wider groove, Minor Groove is the narrow groove in the right-handed B- DNA form

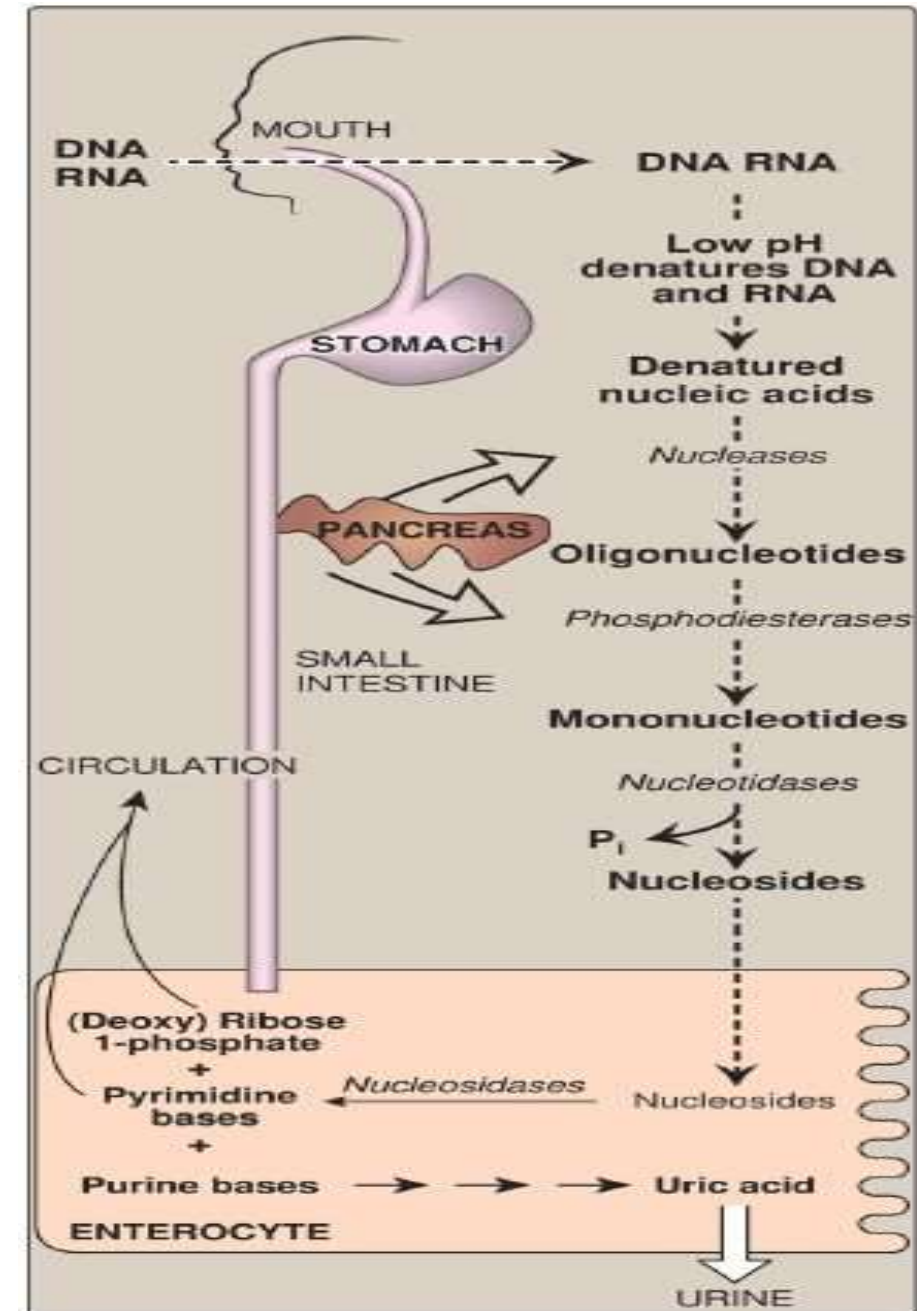
Biosynthesis & Degradation of nucleotides

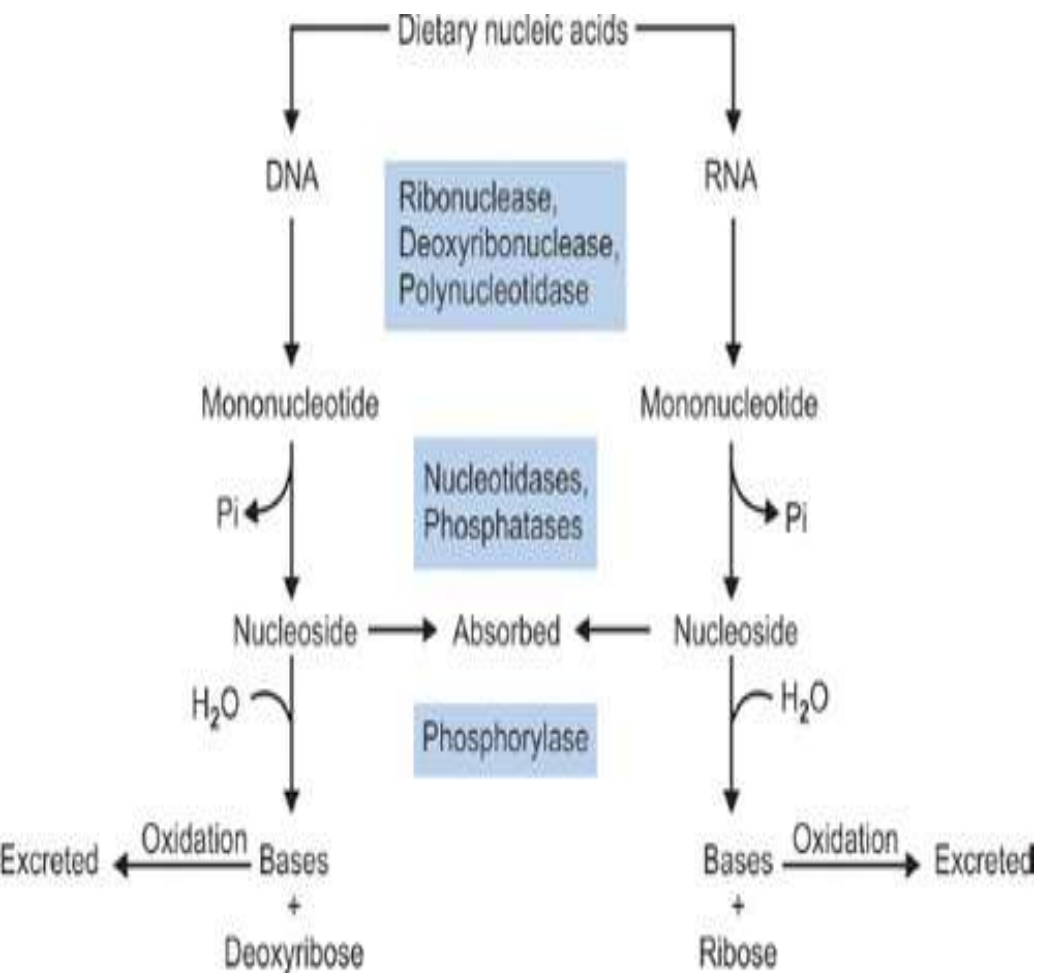
Roles of nucleotides in metabolism

1. Nucleotides are **precursors of nucleic acids** and are needed for:
 - ✓ **replication** of the genome.
 - ✓ **transcription** of the genetic material.
2. **ATP** is the currency of energy for most biochemical pathways.
 - **GTP** also serves as an energy source for certain biological processes.
3. **Nucleotide derivatives**
 - **UDP-glucose** participates in biosynthetic processes ~ the formation of glycogen.
4. Nucleotides are essential components of **signal-transduction pathways**.
 - **cAMP** and **cGMP** are second messengers that transmit signals both within and between cells.
5. ATP acts as the **donor of phosphoryl** groups transferred by **protein kinases**.

Digestion of Nucleic Acids

- Nucleic acids (DNA and RNA) in foods are digested in the small intestine (pancreatic and small intestinal enzymes)
- Pancreatic enzymes called nucleases: ribonuclease (RNA) and deoxyribonuclease (DNA) initiate nucleic acid degradation
- **exonuclease or Phosphodiesterase** (nucleotides removed **from the end**) produce **single nucleotides**
- The two intestinal brush border enzymes (nucleosidase and phosphatase) further break down the nucleotide into pentoses, phosphates, and nitrogenous bases
- The sugar-phosphate can either be reincorporated into nucleotides or metabolized via the Hexose Monophosphate Pathway (R-5-P).





- The purine and pyrimidine bases released are either degraded or salvaged for reincorporation into nucleotides

Salvage pathway	De-Novo synthesis
✓ Free purine and pyrimidines are in large part of salvaged and reused to make nucleotides from degradation of DNA and RNA	✓ Site:- cytosol of the Liver, small intestine, and thymus
✓ Consume less energy than de-novo synthesis and is an energetically preferred source of nucleotide for nucleic acid synthesis	✓ From small molecules readily available in the cell (AA, Ribose-5-Phosphate, CO₂, and NH₃)

Precursors for DNA and RNA

- Both *de novo* and salvage pathways lead to the synthesis of **ribonucleotides**.
- The **deoxyribonucleotides used to build DNA**;
 - are synthesized from the corresponding ribonucleotides.
- The **deoxyribose sugar** is generated by:
 - the **reduction of ribose** within a fully formed nucleotide.
- The **methyl group** that distinguishes the **thymine** of DNA from the **uracil** of RNA is added at the **last step** in the pathway.

Nucleotides Biosynthesis

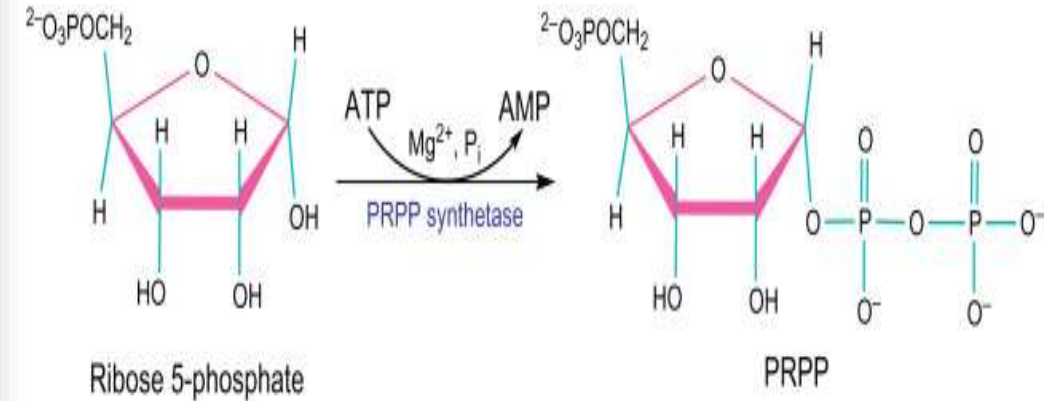
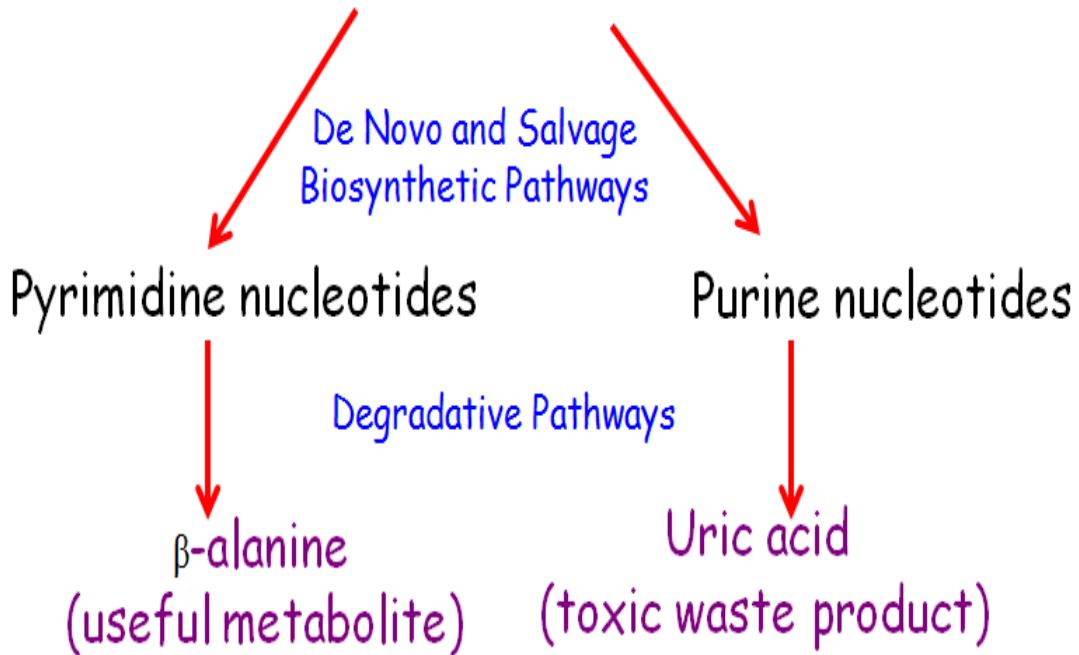
- **Nearly all** organisms synthesize purines and pyrimidines by "*de novo biosynthesis pathway*"
- **Many** organisms also "*salvage*" purines and pyrimidines from **diet and degradative pathways**
- **Ribose generates energy**, but **purine and pyrimidine rings do not**
- **Nucleotide synthesis pathways are good targets for anti-cancer/antibacterial strategies**

Overview of Nucleotide Metabolism

-Biosynthesis-

(5-phosphoribosyl- α -pyrophosphate)

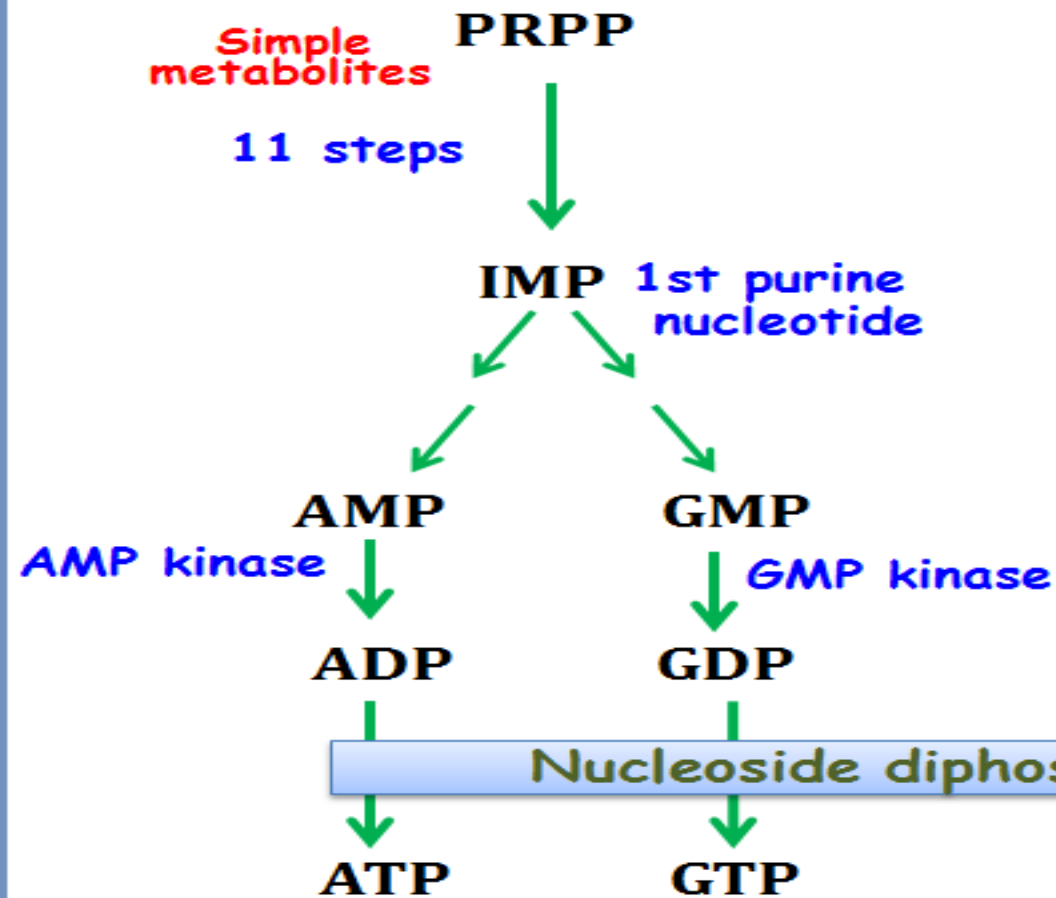
PRPP



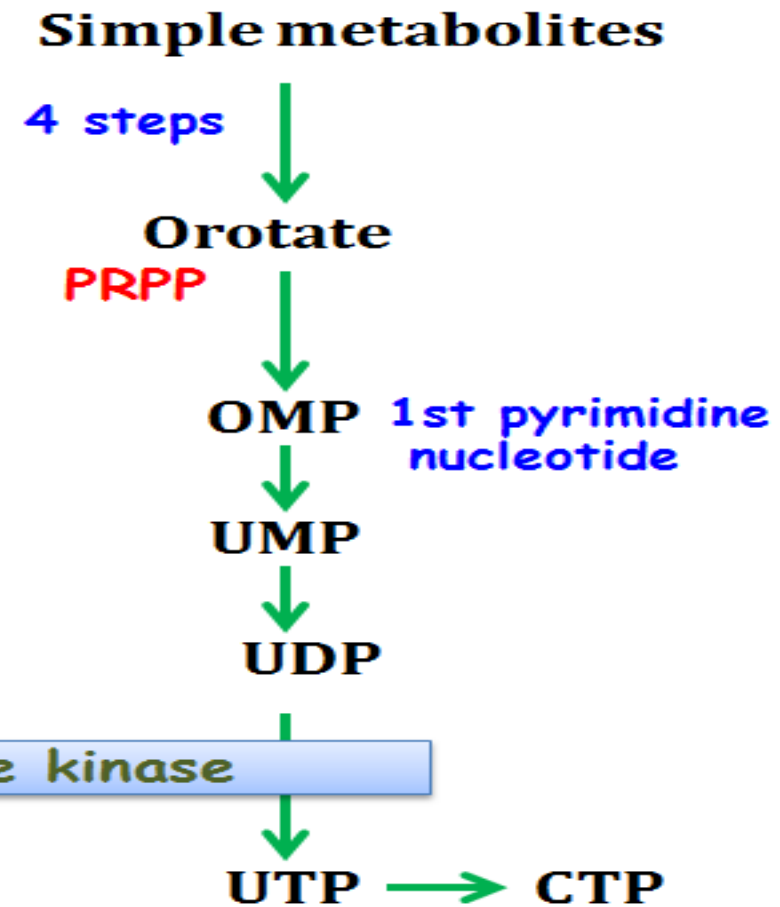
PRPP is activated form of R-5P, and used for the **formation** of purine nucleotides via inosine-5-monophosphate (IMP)

Continue....

Purines

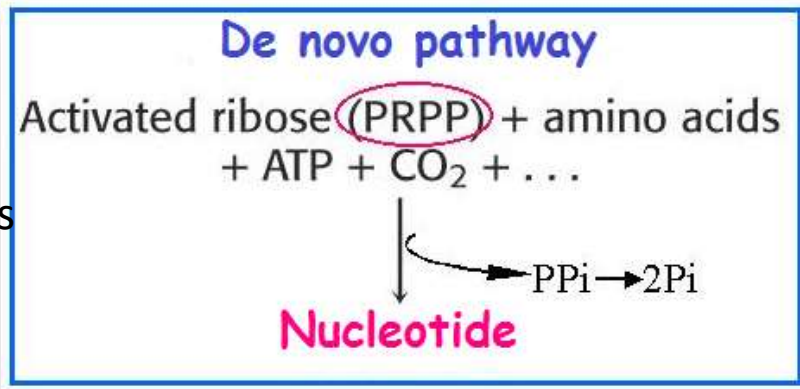


Pyrimidines



Nucleotide Biosynthesis

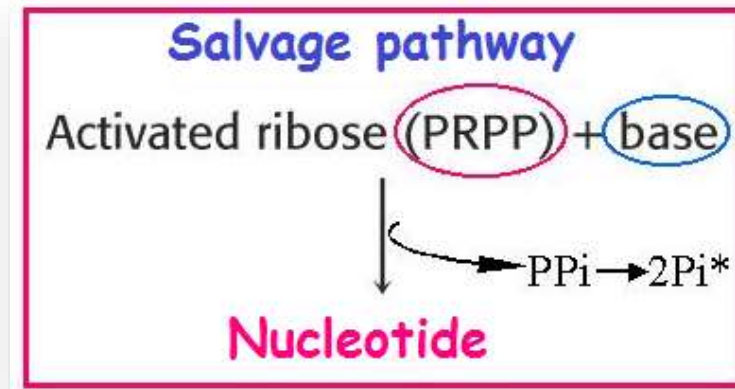
the **de novo** pathway enzymes **build purine** and **pyrimidine** nucleotides from “**scratch**” using simple molecules such as



Begins with their **metabolic precursors:**

✓ amino acids, ribose 5-phosphate, CO₂, and NH₃.

✓ ATP hydrolysis is required for de novo synthesis.



Some tissue (brain) are salvage dependent

Recycles the free bases and nucleosides released from **nucleic acid** breakdown.



Question: Why do we need both pathways?

In these pathway, two features deserve to mention:

- **First**, there is evidence, especially in the *de novo* purine pathway
 - ✓ the enzymes are present as **large, multienzyme complexes** in the cell
- **Second**, the cellular pools of nucleotides (other than ATP) are **quite small**,
 - **1% or less of the amounts** required to synthesize the cell's DNA
- **Therefore:**
 - **cells must continue to synthesize nucleotides**, and
 - in some cases nucleotide synthesis **may limit the rates of DNA replication and transcription**

Because of the **importance of these processes in dividing cells**,
agents that inhibit nucleotide synthesis have become
particularly **important to modern medicine**.

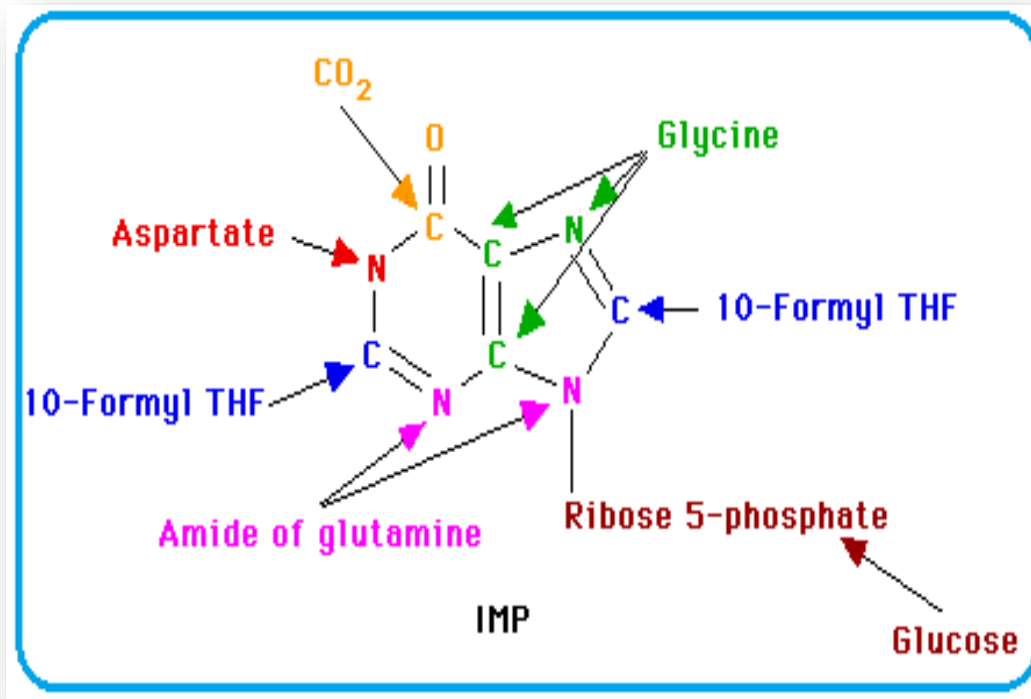


Purine Biosynthesis

- The purine bases are produced by *de novo* pathways:
 - ✓ uses **amino acids**, CO_2 , and N^{10} -formyl tetrahydrofolate as **precursors** to produce nucleotides
- Most *de novo* synthesis occurs:
 - ✓ in the **liver**, and
 - ✓ the **nitrogenous bases** and **nucleosides** are then transported to other tissues by **red blood cells**
- The **brain** also synthesizes **significant amounts of nucleotides**
- B/c **de novo pathway** requires:
 - ✓ **More than six high-energy bonds** per purine produced
- **A salvage pathway used by many cell types**,
 - ✓ can convert **free bases** and **nucleosides** to **nucleotides**

De novo Purine Nucleotide Biosynthesis

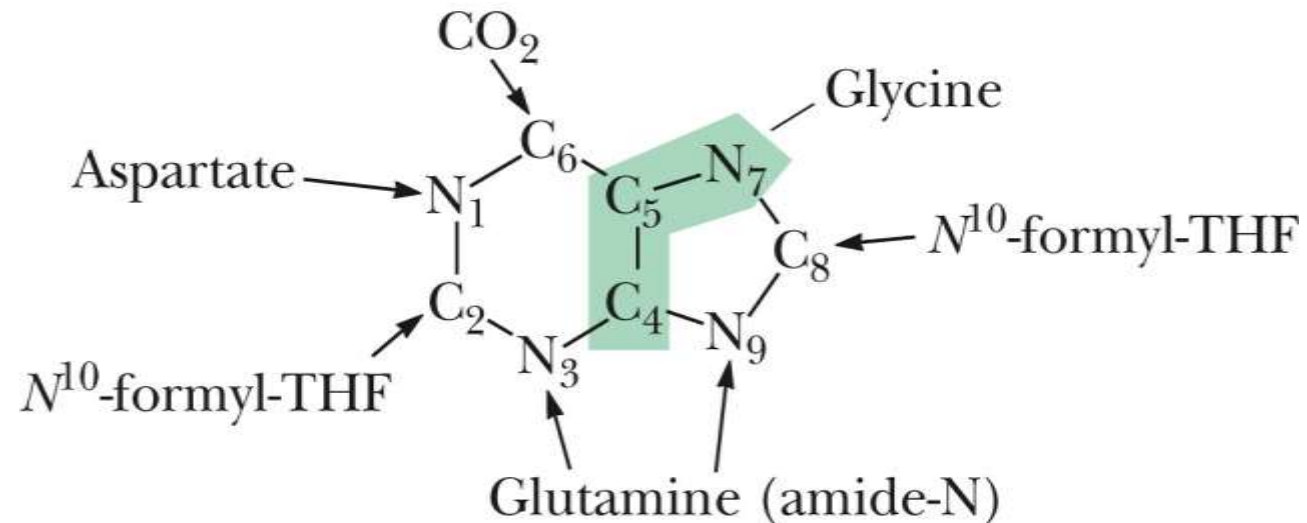
- The two parent purine nucleotides of nucleic acids are:
 - ✓ adenosine 5-monophosphate (AMP) and
 - ✓ guanosine 5-monophosphate (GMP).
- The **origin** of the **carbon and nitrogen atoms** of the **purine ring system**, as determined by **John Buchanan** using isotopic tracer experiments in birds



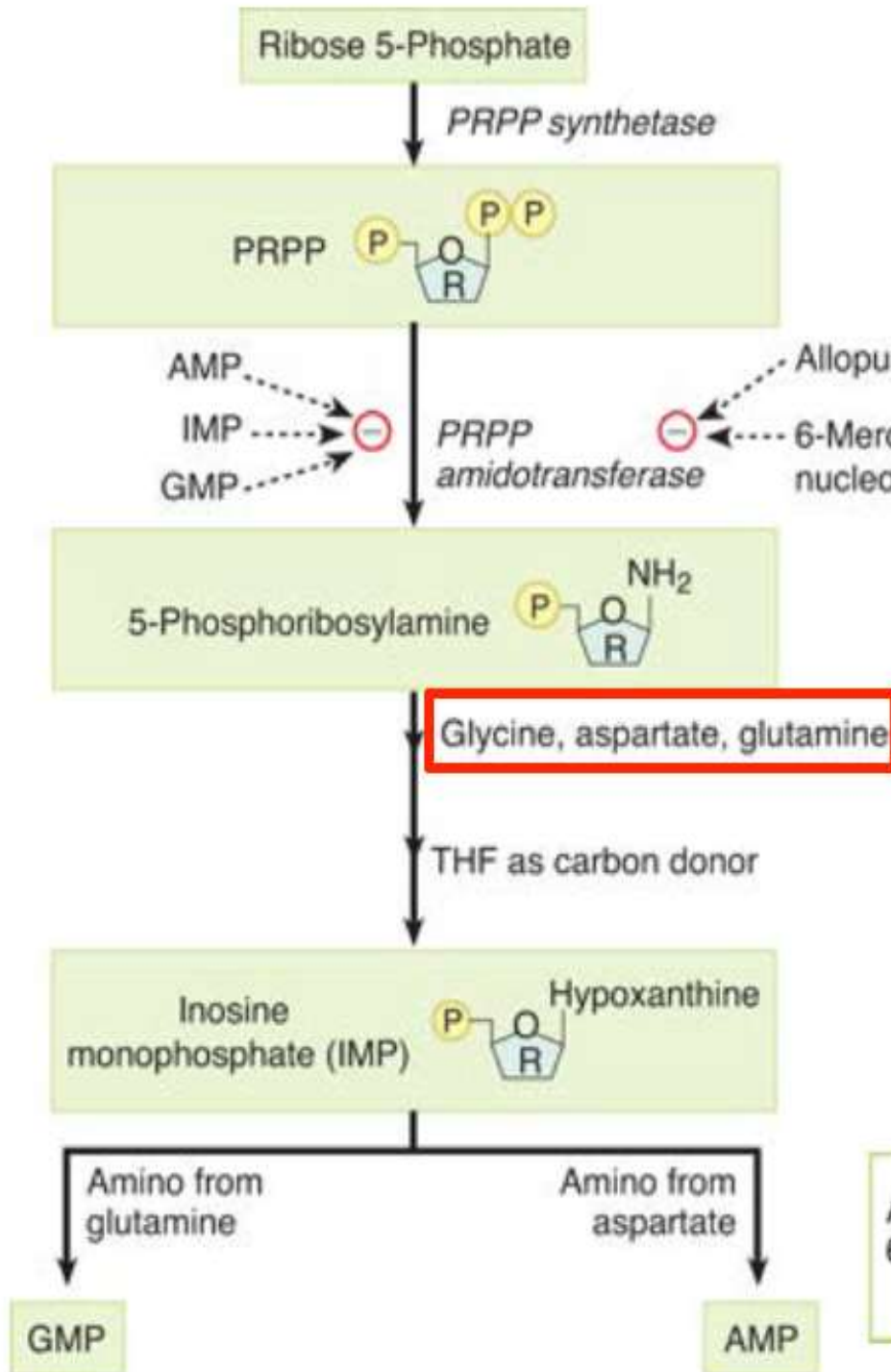
How Do Cells Synthesize Purines?

▪ *John Buchanan (1948) "traced" the sources of all nine atoms of the purine ring*

- N-1: aspartic acid
- N-3, N-9: glutamine (amide-N)
- C-2, C-8: N^{10} -formyl-THF - one carbon units
- C-4, C-5, N-7: glycine
- C-6: CO_2



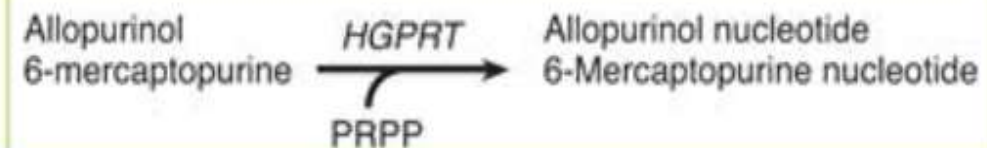
Purine Synthesis Pathway



End product is feedback inhibition (binding to a different site) to a key anabolic enzyme (PRPP amidotransferase)

drug **6-mercaptopurine (6-MP)** → **Anticancer** (acute lymphoblastic leukemia) "**inhibits de novo purine synthesis**" and acts as an "**antiproliferative agent**" by interfering with **protein, DNA, and RNA synthesis** and **promoting apoptosis**.

Allopurinol (in high concentration) "**inhibits de novo purine synthesis**" in the **absence of xanthine oxidase**. (for uric acid)



- Step 1: Ribose-5-phosphate pyrophosphokinase.
- Step 2: Glutamine phosphoribosyl pyrophosphate amidotransferase.
- Step 3: Glycinamide ribonucleotide (GAR) synthetase.
- Step 4: GAR transformylase.
- Step 5: FGAM synthetase (FGAR amidotransferase).
- Step 6: FGAM cyclase (AIR synthetase).
- Step 7: AIR carboxylase.
- Step 8: SAICAR synthetase.
- Step 9: adenylosuccinase.
- Step 10: AICAR transformylase.
- Step 11: IMP synthase.

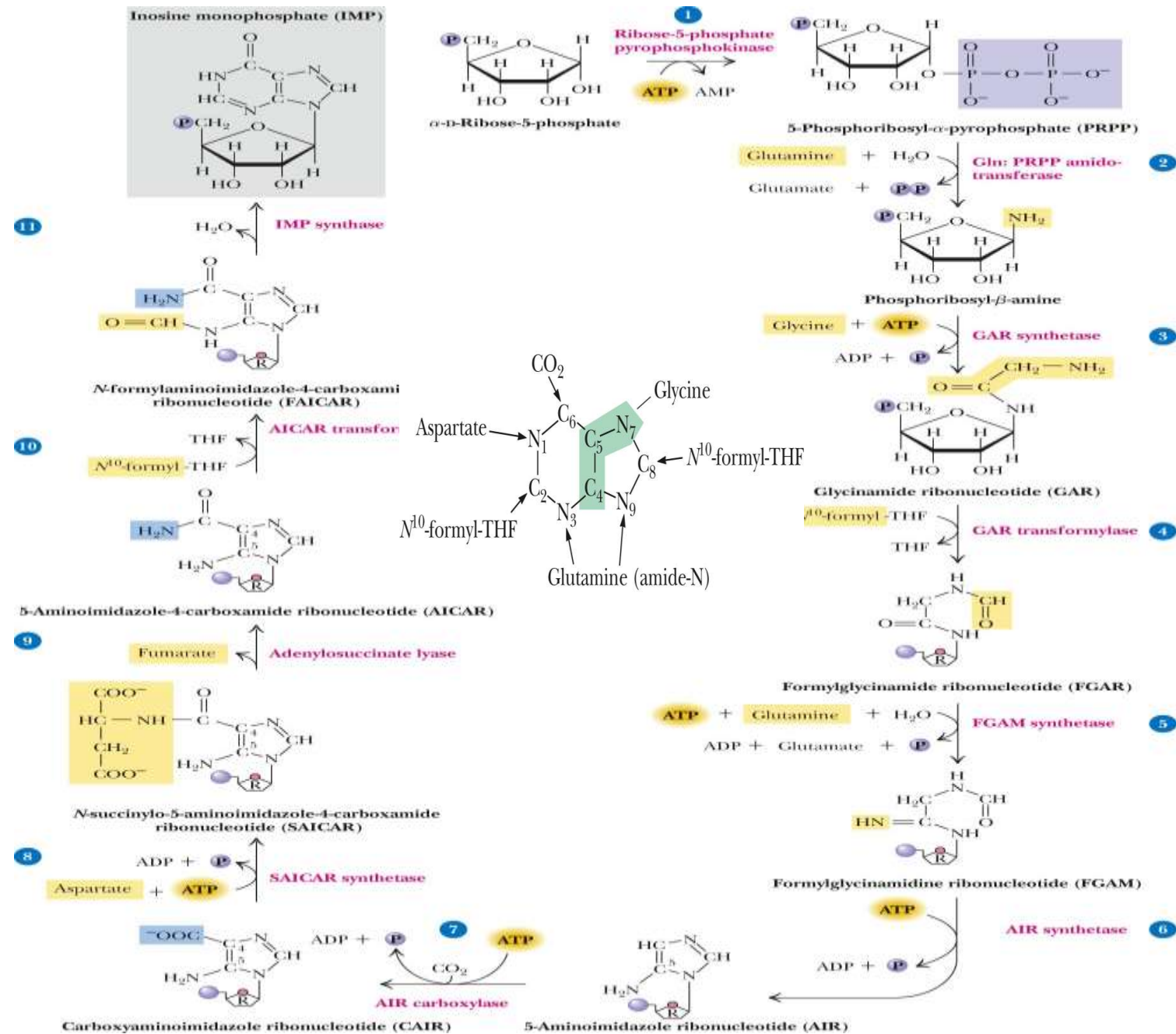
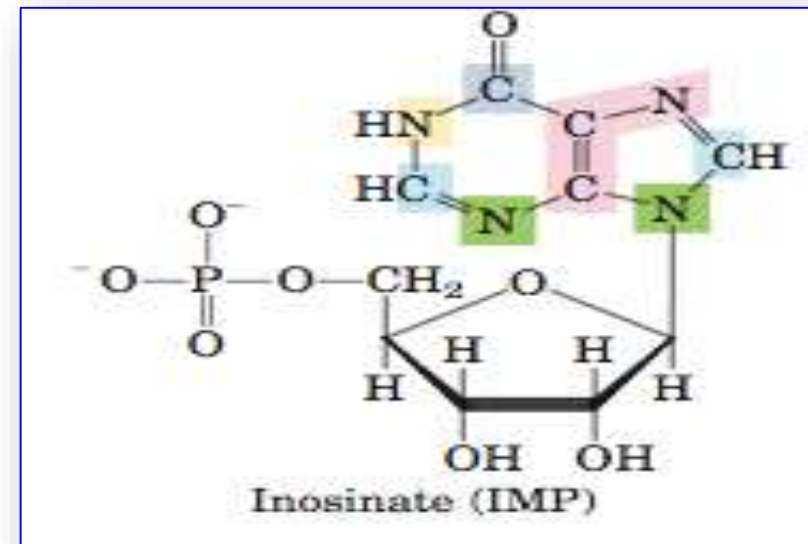


Figure
The *de novo* pathway for purine synthesis

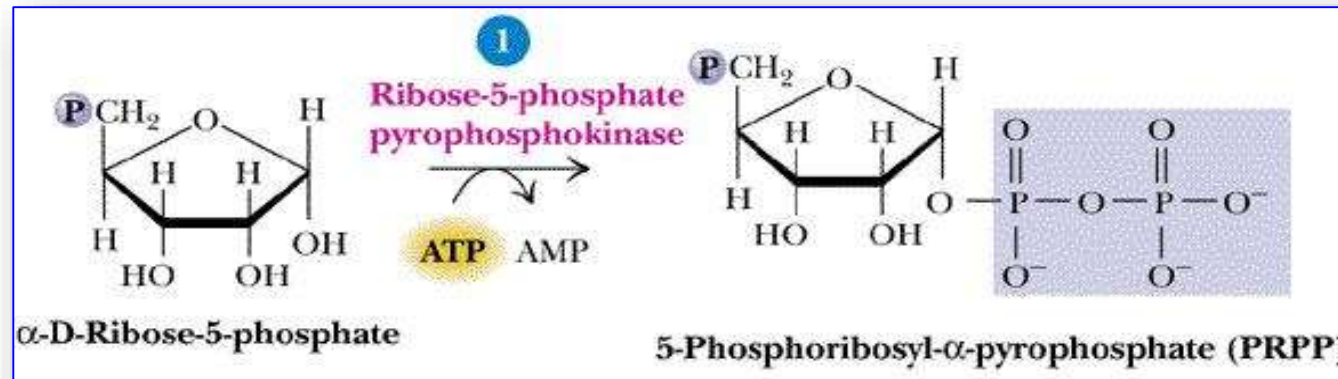
Synthesis of Inosine Monophosphate (IMP)

- Basic pathway for biosynthesis of **purine ribonucleotides**
- Starts from **ribose-5-phosphate** which is derived from the **pentose phosphate pathway**
- Requires **11 steps overall**
- Occurs primarily in the **liver**

IMP is the intracellular intermediate precursor for both (**AMP**) and (**GMP**) and thus plays a “central role in intracellular purine metabolism”



Step 1: Activation of ribose-5-phosphate



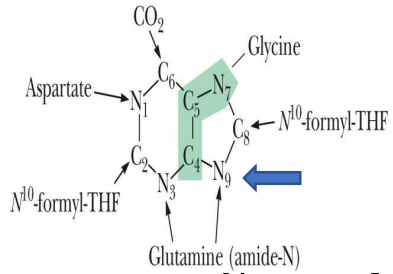
R-5-P from PPP needs to be activated 1st to cyclic PRPP (2 ATP requiring step) contains **2 phosphates**.

- **Enzyme:** Ribose-5-phosphate pyrophosphokinase
- **Product:** 5-phosphoribosyl- α -pyrophosphate (PRPP).
- This step is **tightly regulated by feedback inhibition**.
- PRPP is also a **precursor** in the biosynthesis of:
 - ✓ **pyrimidine** nucleotides and
 - ✓ the amino acids **histidine** & **tryptophan**.

The enzyme has an “**allosteric regulation site**”, activated by phosphate (substrate) and inhibited by ADP(low energy), both **compete** for the **allosteric site**

Step 2: Acquisition of purine atom 9

What is the properties of committed step



Second step: **Gln PRPP amidotransferase**

Form **phosphoribosyl- β -amine**;

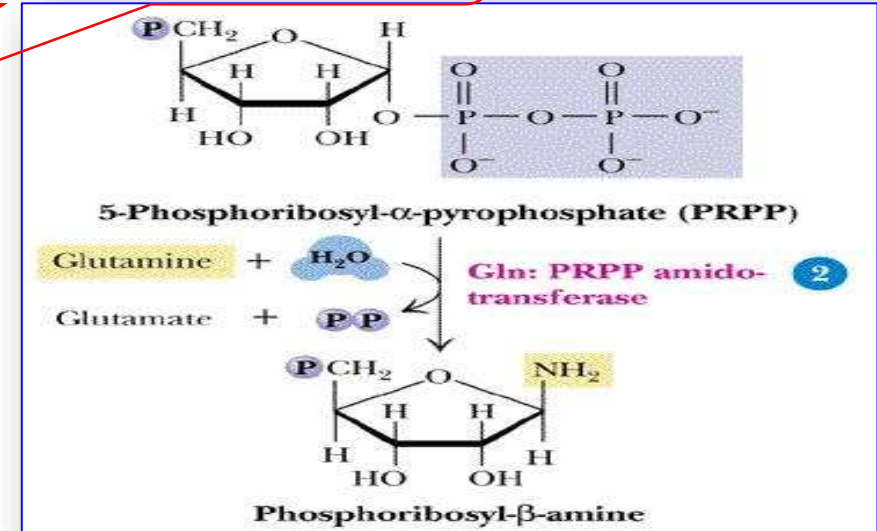
Changes C-1 configuration ($\alpha \rightarrow \beta$)

GMP and AMP inhibit this step - but at **distinct** sites

Azaserine - Gln analog -inhibitor/ **anti-tumor**

- Inhibit by **glutamine analogs**: azaserine and acivadin
- **Suicide inactivators**

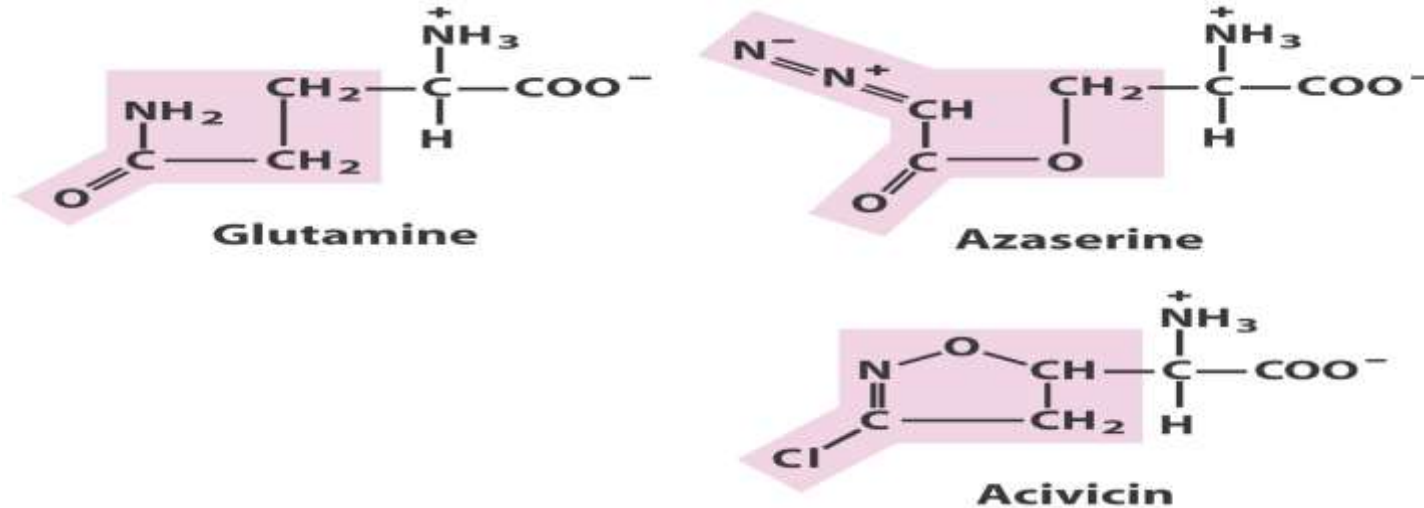
- In this **committed step**, an amino group donated by **glutamine** is attached at C-1 of PRPP.
- **Enzyme**: **Glutamine phosphoribosyl pyrophosphate amidotransferase**.
- **Product**: Resulting in the formation of **5-phosphoribosyl- β -amine**.



- PRPP is a branch point so this step is a committed step (highly regulated),
- Gln PRPP amidotransferase → is an allosteric enzyme regulated by AMP and GMP (allosteric regulation) feedback Inhibition
- PRPP itself → “feed-forward allosteric activation” to *Gln PRPP amidotransferase*

Cancer Chemotherapy:

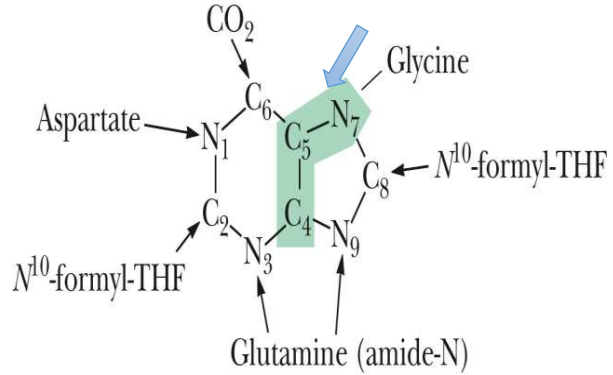
- Acts as an **irreversible inhibitor** of **glutamine-dependent enzymes**
 - ✓ by **covalently attaching** to nucleophilic groups in the **glutamine-binding site**.



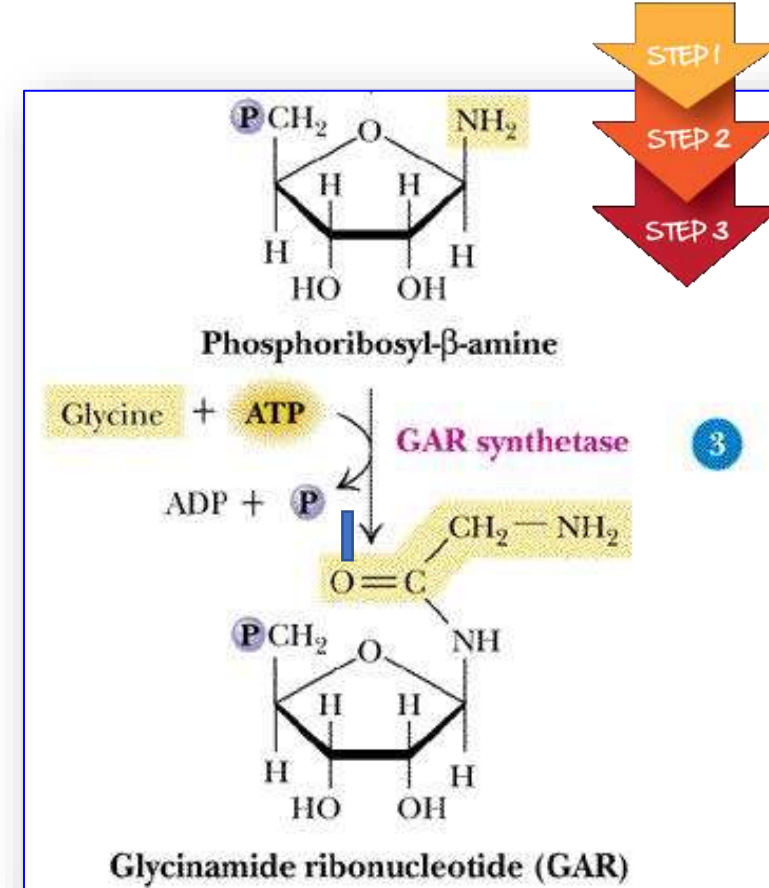
→ **Azaserine**” anti-cancer, functions as a **purine antagonist** and **glutamine analog** that **competitively inhibits** pathways in which **glutamine is metabolized** (With **no formation** of product)

→ **Glutamine analog** → **Gln PRPP amidotransferase inhibitor**
“anti-cancer antibiotics” **irreversibly bind** to the **active site** .
→ Classical **cytotoxic immunosuppressants** act by inhibiting **DNA synthesis**. Suppress immune function.
→ Making the **enzyme site inaccessible** to **glutamine** (no addition of amino group) → leads to inhibition of the step.

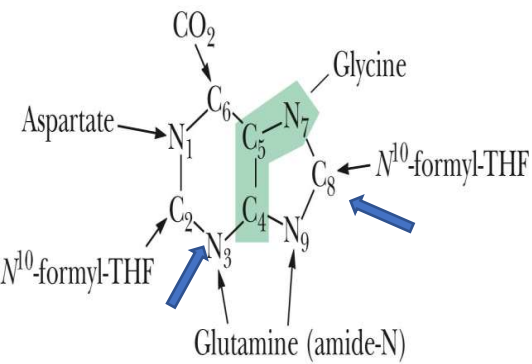
Step 3: Acquisition of purine atoms C4, C5, and N7



Synthetase “requires ATP” but **not synthase** (used in the synthesis of Bio mol **without energy**)



- Phosphoribosylamine reacts with **ATP** and **glycine** to produce **glycinamide ribonucleotide (GAR)**
- **Enzyme: GAR synthetase**



Step 4: Acquisition of purine atom C8

- Formylation of a free α -amino group of GAR.
- **Enzyme: GAR transformylase.**
- **Co-factor of enzyme: N¹⁰-formyl THF**

Step 5: Acquisition of purine atom N3

- The **amide amino group** of a second **glutamine** is transferred to form formylglycinamidine ribonucleotide (**FGAM**).
- The reaction is driven by the hydrolysis of **ATP**.
- **Enzyme: FGAM synthetase.**
- **Irreversibly inactivated** by **azaserine**

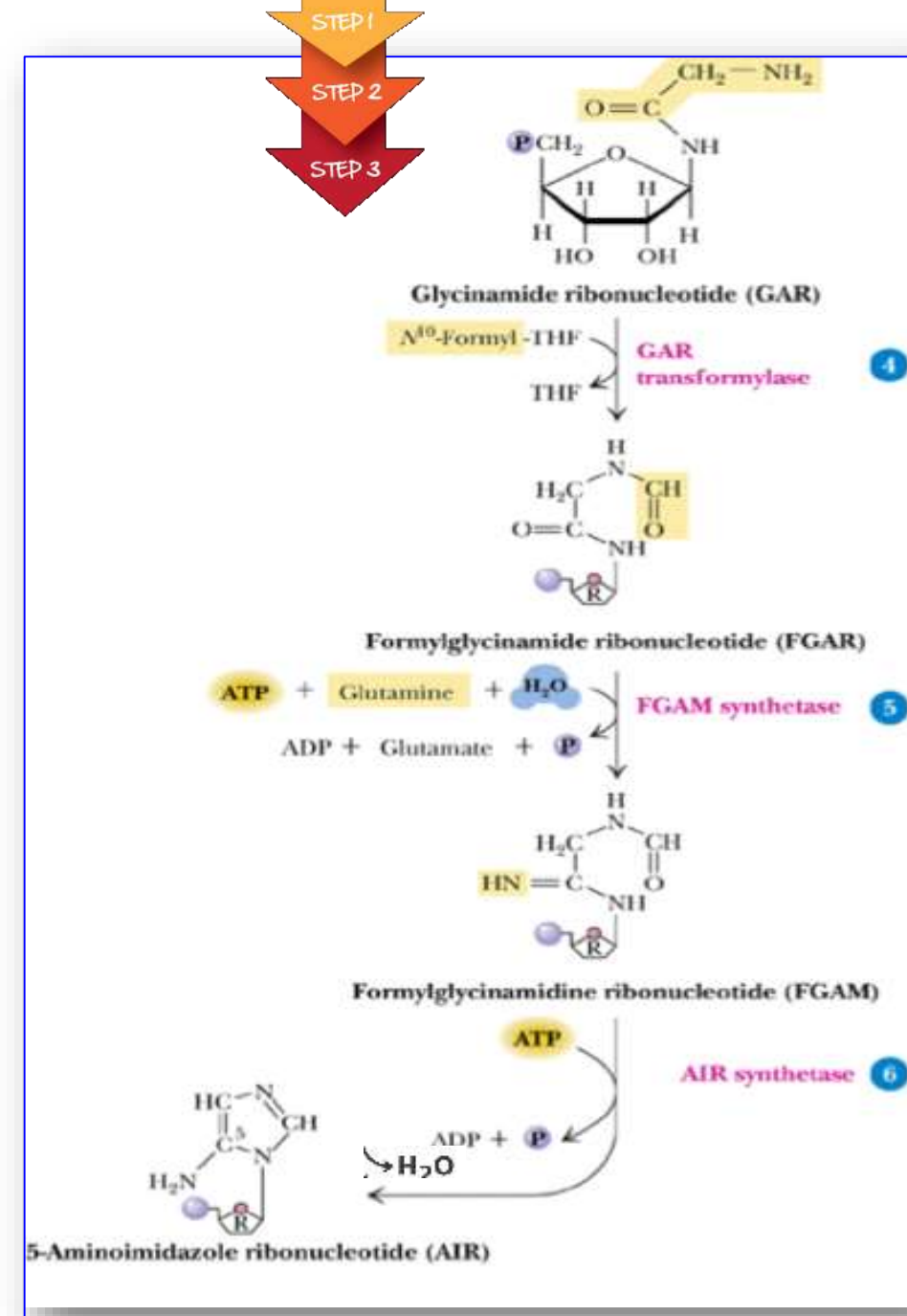
Step 6: Closing the ring

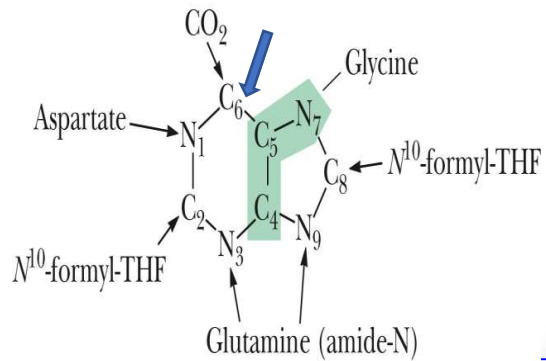
- Closing of the imidazole ring or formation of **5-aminoimidazole ribonucleotide (AIR)**.
- The reaction is driven by the hydrolysis of **ATP**.
- **Enzyme: AIR synthetase.**

THF → source is **folic acid**
(**Vit B9**) containing **diet**

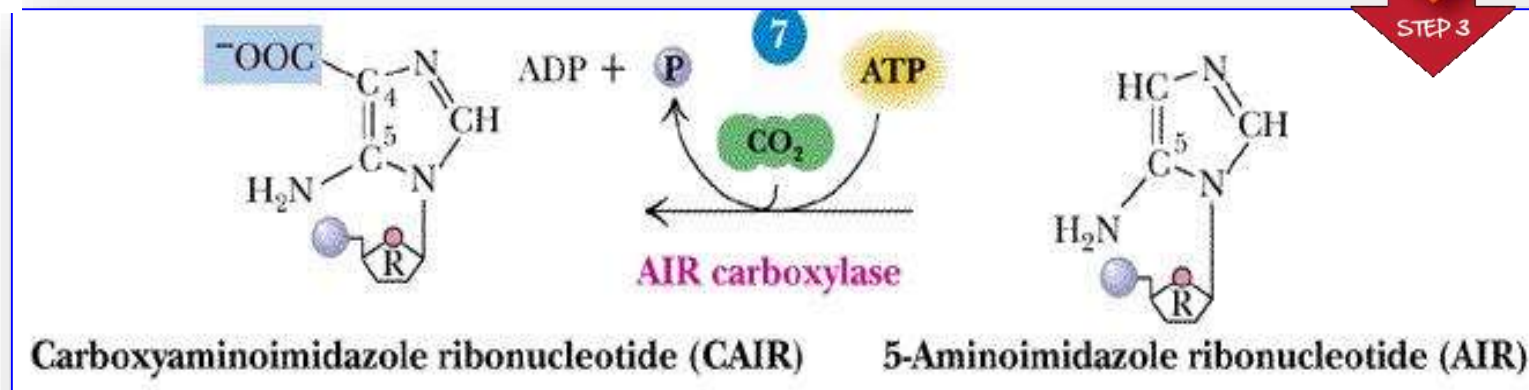
F.A → DHF → THF (in the human body) by **THF reductase**, not bacteria
(Can synthesize it)

STEP 5 → “Azaserine
(**anti-cancer drug**)” can **Inhibit FGAM synthetase**(step 5) similar to step 2, **B/c its glutamine analog**





Step 7: Acquisition of C6



- C6 is introduced as HCO_3^-
- The reaction is driven by the hydrolysis of **ATP**
- **Enzyme:** **AIR carboxylase** (aminoimidazole ribonucleotide carboxylase)
- **Product:** **CAIR** (carboxyaminoimidazole ribonucleotide)

Step 8: Acquisition of N1

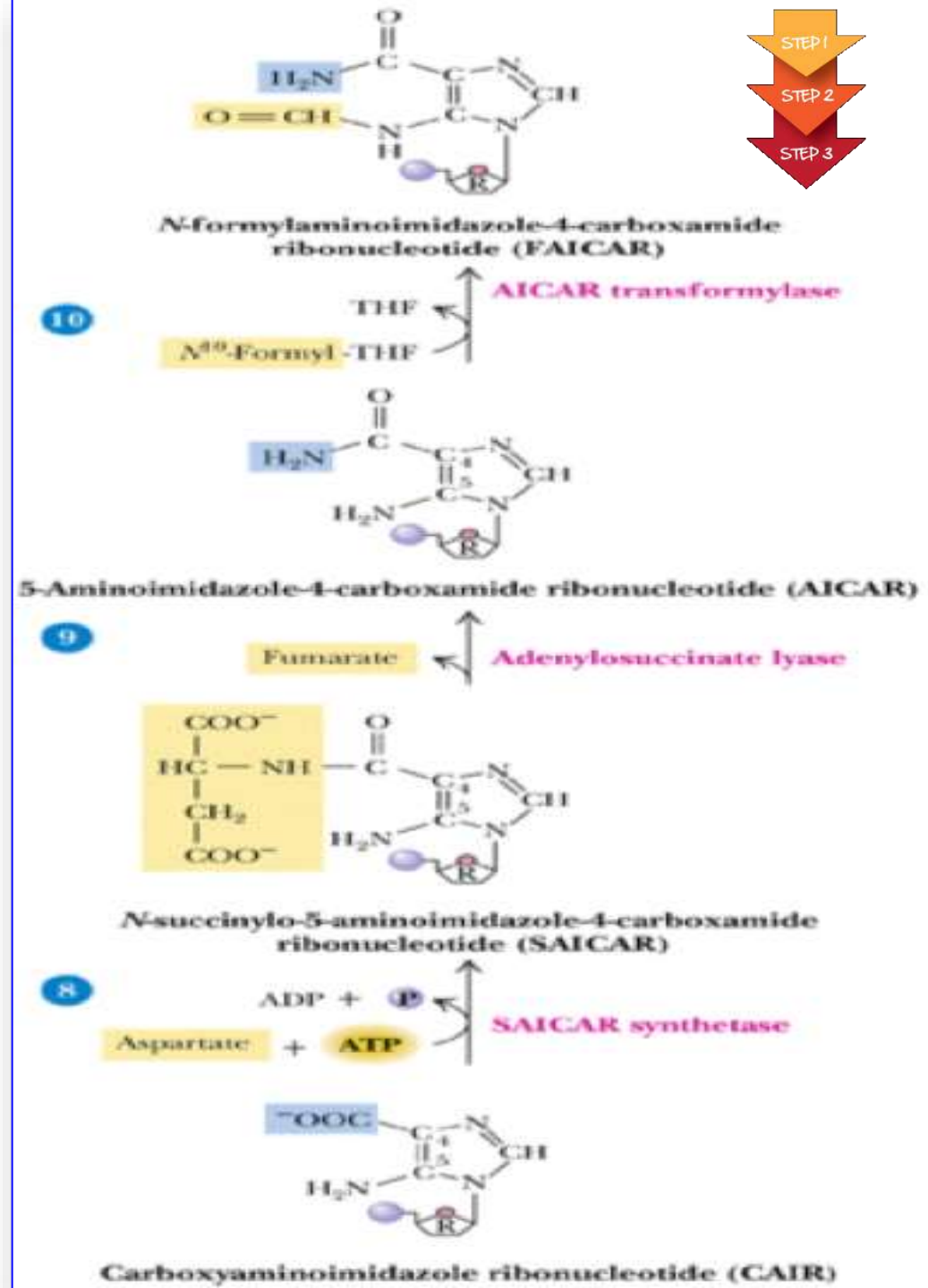
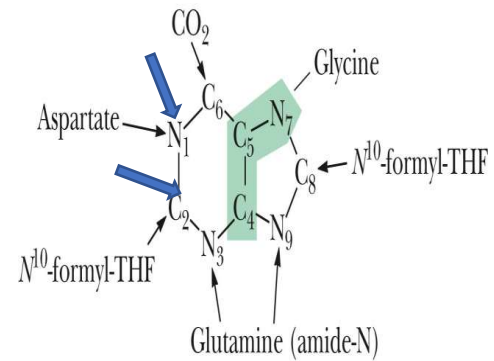
- N1 is acquired from **aspartate** in an amide condensation reaction that is driven by hydrolysis of **ATP** to produce N-succinyl-5-aminoimidazole-4-carboxamide ribonucleotide (SAICAR).
- Enzyme: SAICAR synthetase**

Step 9:

- Elimination of fumarate** by the action of **adenylosuccinate lyase** to produce 5-aminoimidazole-4-carboxamide ribonucleotide (AICAR)

Step 10: Acquisition of C2

- Another formylation reaction catalyzed by **AICAR transformylase** results in the formation of 5-formylaminoimidazole-4-carboxamide ribonucleotide (FAICAR)
- incorporation of a **formyl group** from **N10 formyl THF**
- Anti-cancer and anti-bacterial drug** work by inhibiting THF formation (inhibit this step)



Biochemistry in medicine



- The dependence of **purine biosynthesis** on **folic acid** compounds at Steps 4 and 10 means that:
 - ✓ **antagonists of folic acid metabolism indirectly inhibit purine formation** and,
 - ✓ in turn, ***inhibit nucleic acid synthesis, cell growth, and cell division***
- Clearly, **rapidly dividing cells** such as
 - ✓ **Malignancies (fast-growing tumors) or infective bacteria** are **more susceptible to these antagonists** than slower-growing normal cells
- The dependence of purine biosynthesis on **THF** in two steps means that **methotrexate** and **sulfonamides block purine synthesis**

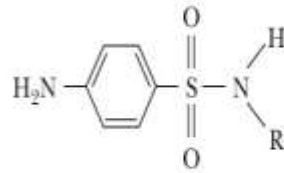
- **Folic acid (THF is active form)** → is **vitamin B9 water soluble**, THF is used for the **maturation of RBC** folate **deficiency** causes **megaloblastic anemia**, the source is **green vegetables, Lentin, oranges, cereal, beans, and the liver..** body needs it in a small amount to function
- **Cancer cell and bacteria** has a **faster rate of purine biosynthesis** than normal cell
- **“THF analog”** or **metabolism antagonist drug** is used for the treatment

Folate Analogs as Antimicrobial and Anticancer Agents

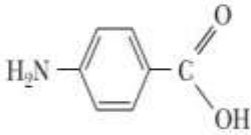
- *De novo* purine biosynthesis depends on **folic acid compounds** at steps 4 and 10
- **Sulfonamides** are effective anti-bacterial agents
- **Methotrexate and related compounds** **inhibit** the **reduction of dihydrofolate to tetrahydrofolate**, catalyzed by **dihydrofolate reductase**.
- **Methotrexate** and **Aminopterin** are folic acid analogs that have been used in **cancer chemotherapy** but are also toxic to all dividing cells
- **Methotrexate chemotherapy** is a **Competitive inhibitor of DHFR (folic acid analog) folate antagonist** → and therefore interferes with DNA synthesis, repair, and cellular replication. **Inhibit the 2 steps.**

(a)

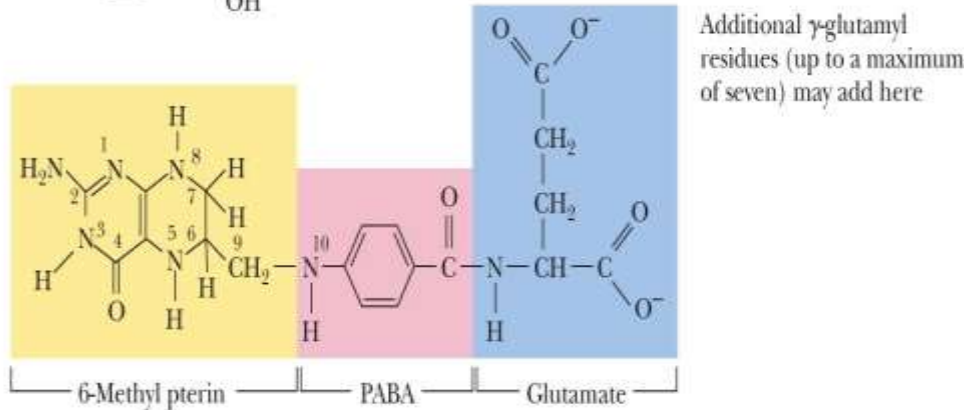
Sulfonamides have the generic structure:



PABA (*p*-aminobenzoic acid)



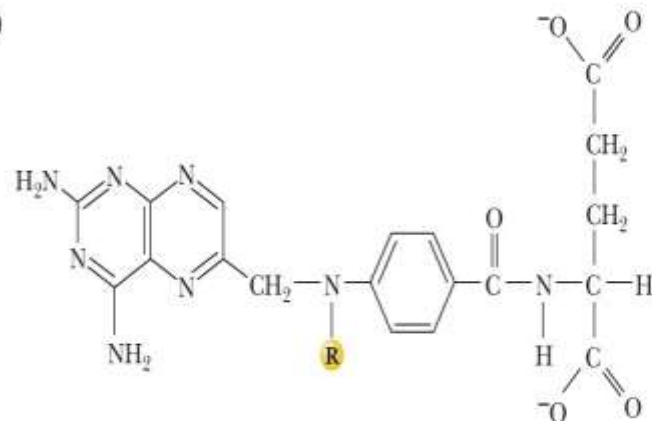
THF (tetrahydrofolate)



Sulfa drugs, or sulfonamides

- their antibiotic properties are due to **their similar to *p*-aminobenzoate (PABA)**, an important precursor in the folic acid synthesis
- Sulfonamides block folic acid formation by competing with PABA**

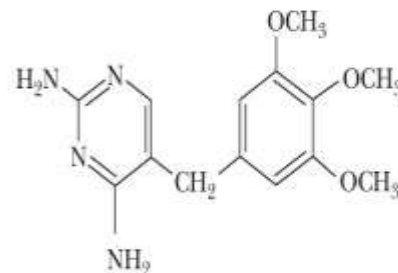
(b)



2-Amino, 4-amino analogs of folic acid

R = H Aminopterin

R = CH₃ Amethopterin (methotrexate)



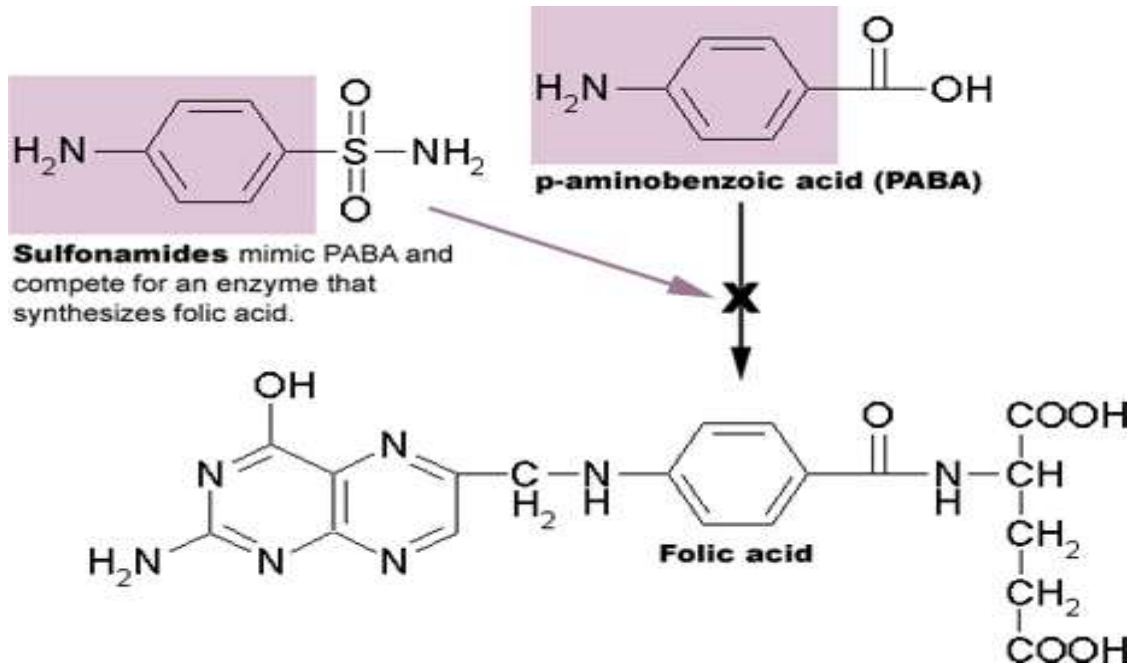
Trimethoprim

Sulfa drugs are classified as **anti-metabolite** antibiotics,

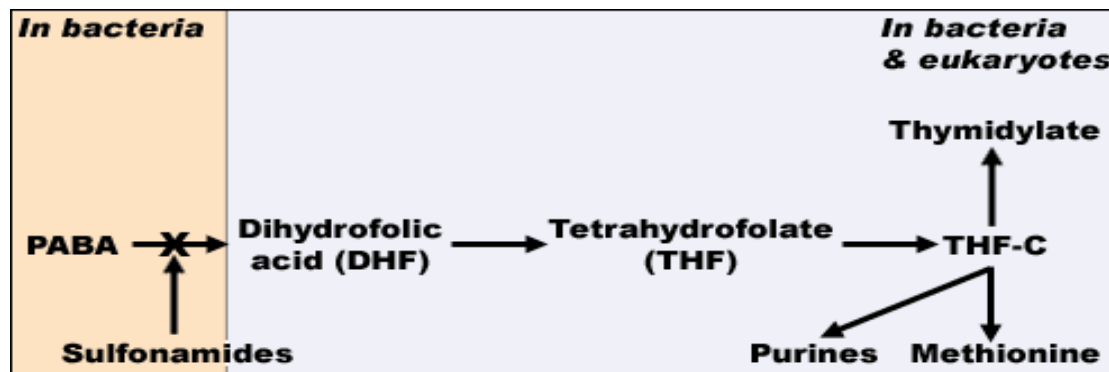
Trimethoprim is anti-bacterial.

Anti-cancer chemotherapy (antimetabolite) → binds to dihydrofolate reductase with about **1000-fold greater affinity** than DHF and thus acts as a virtually irreversible inhibitor.

- **p-aminobenzoic acid (PABA)** is required by many bacteria to produce an important enzyme cofactor, **THF**
- These bacteria require THF for their growth and division

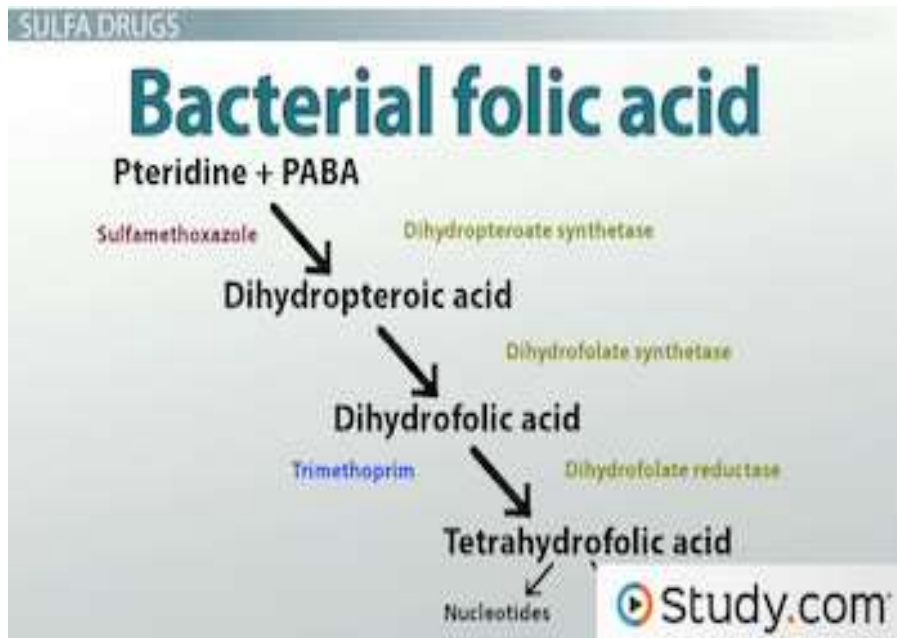


- **Sulfanilamide** acts as a **competitive inhibitor** to enzymes that convert PABA into folic acid, resulting in a depletion of this cofactor.
- The depletion of this cofactor results in retarded growth and eventual **death of the bacteria**.

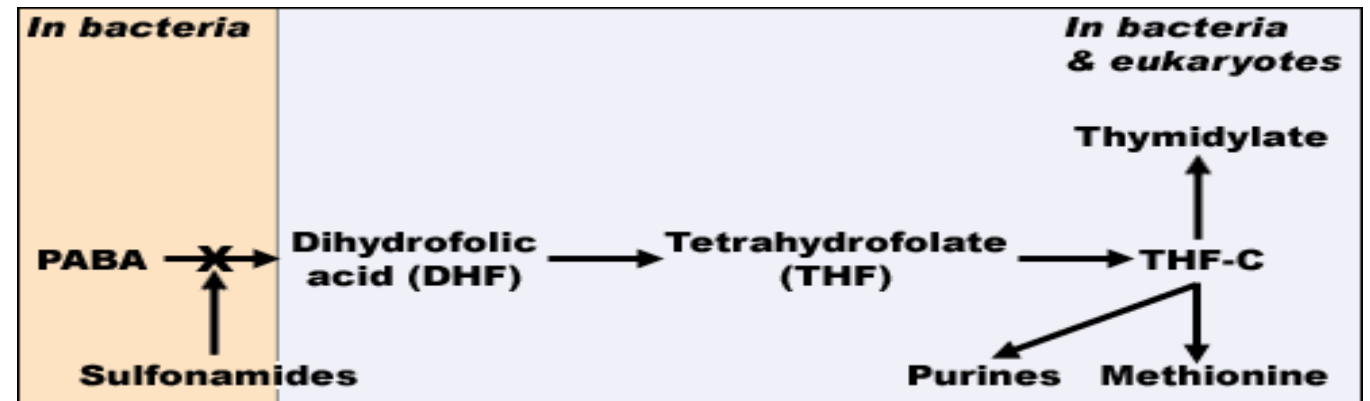


- Mammals absorb their folic acid from their diets, so **sulfanilamide** exerts **no effects** on them.

Sulfonamide selectively kills bacteria and is effective “without affecting human cells” because the folic acid source is food



- Vitamin B9 cannot enter bacterial cells and thus bacteria must make their own from **PABA** and **pteridine**



Sulfanilamide Compete for dihydropteroate synthase

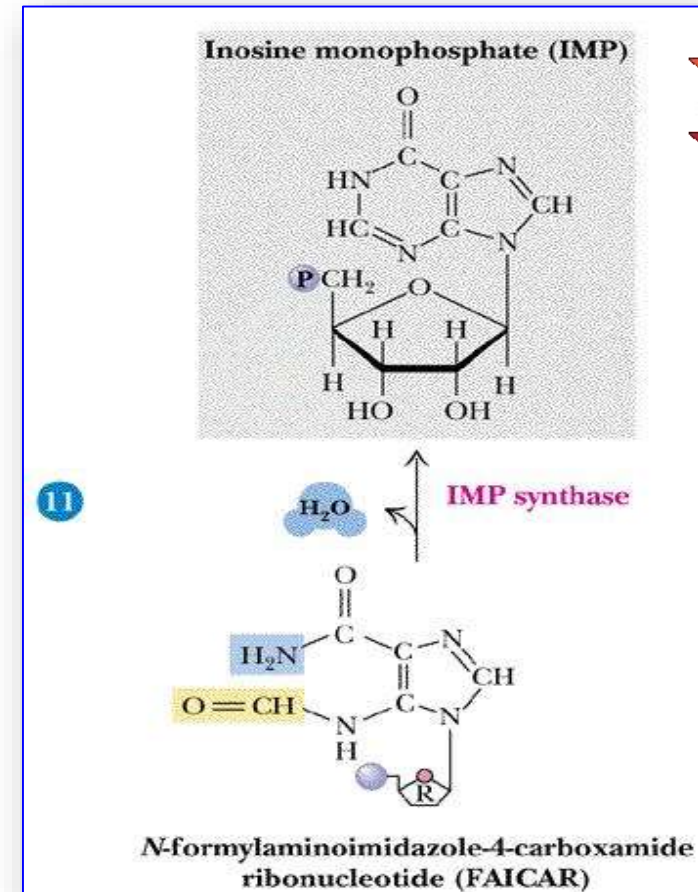
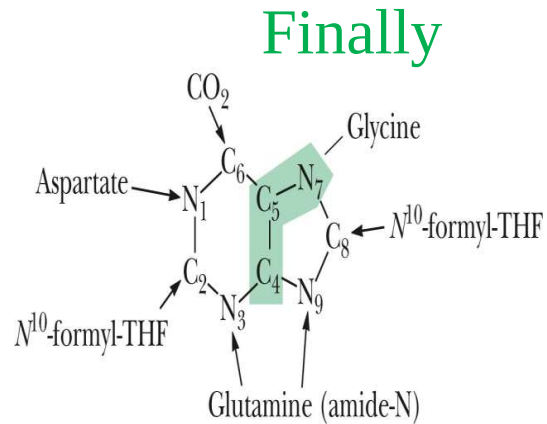
Bacteria: PABA → DHF → THF

Mammal: Folic acid (diet) → DHF → THF

...cont'd

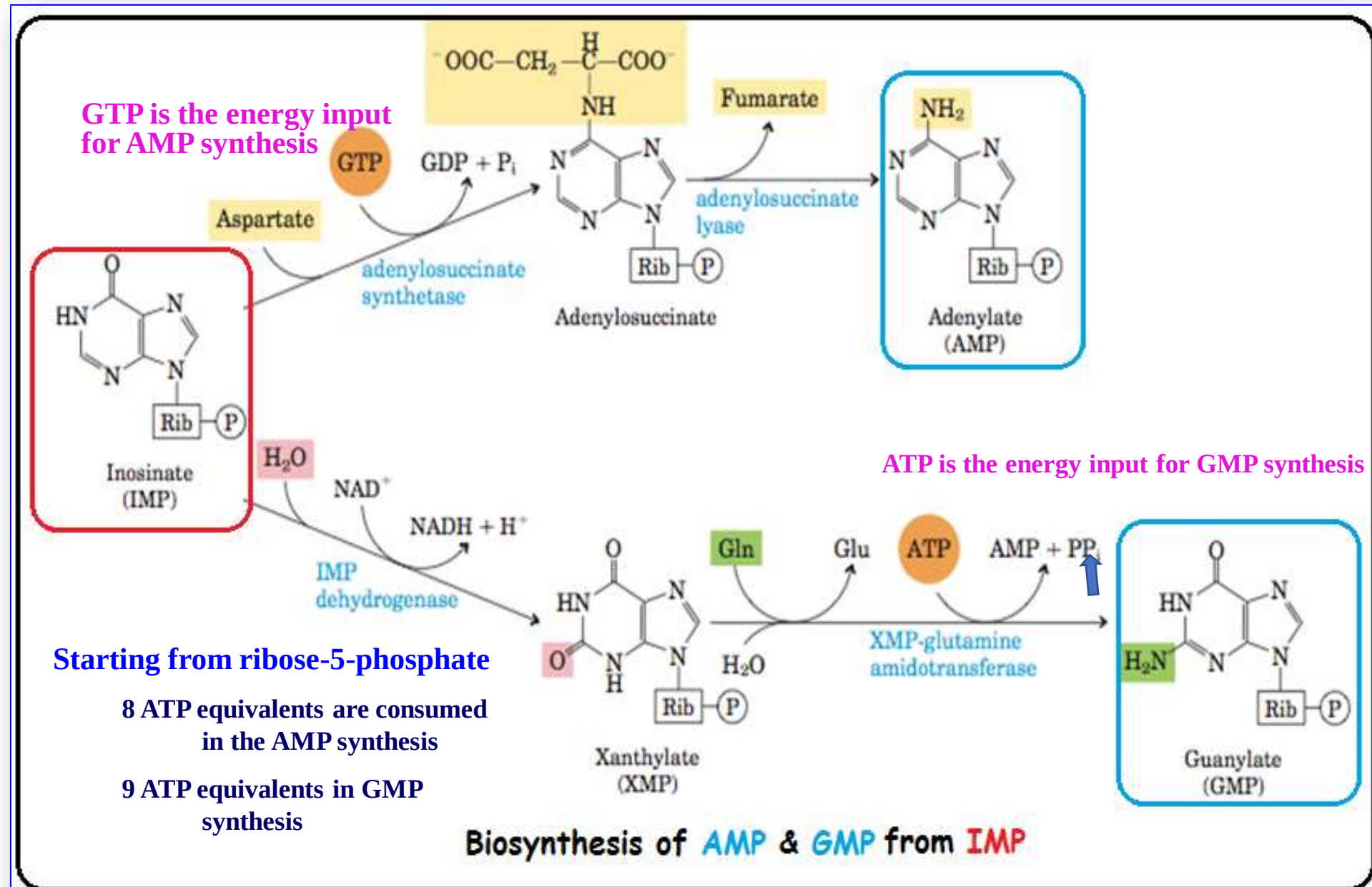
- Inhibitors of human purine synthesis are **extremely toxic to tissues**,
 - especially to developing structures such as in a fetus, or
 - to cell types that normally replicate rapidly,
 - Most of the chemo **side effects**... has **toxicity to normal dividing cell**
 - including those of bone marrow, skin, gastrointestinal (GI) tract, immune system, or hair follicles.
- As a result, individuals taking such **anticancer drugs can experience adverse effects, including**
 - **anemia,**
 - **scaly skin,**
 - **GI tract disturbance,**
 - **Immuno-deficiencies, and**
 - **hair loss.**

Step 11: Cyclization or ring closure

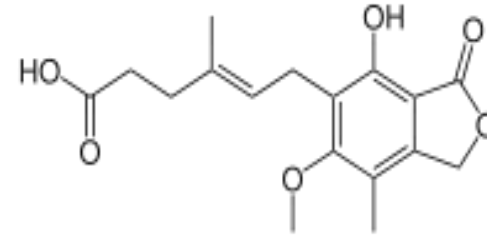


- **Water** is eliminated by the action of **inosine monophosphate (IMP) synthase**
- In contrast to step 6 (closure of the imidazole ring), this reaction **does not require ATP hydrolysis**
- Once IMP is formed, it is **rapidly converted to AMP and GMP**

Biosynthesis of AMP and GMP from IMP



Mycophenolic Acid



- **Inhibits** *inosine monophosphate dehydrogenase*
- Enzyme converts **IMP to xanthosine monophosphate** (to be converted to **GMP**)
- **B and T lymphocytes** need nucleotides to divide
- **Drug** is used to **prevent organ transplant graft rejection**
- **Mycophenolic Acid** is an antineoplastic antibiotic (immunosuppressant medication) derived from various **Penicillium** fungal species. Used during transplant, and prevent autoimmune (lupus)
- MPA, inhibits (IMPDH), preventing the formation of **GMP** and “synthesis of lymphocyte DNA” which results in the inhibition of lymphocyte proliferation, antibody production, cellular adhesion, and migration of T and B lymphocytes. To prevent rejection
- in patients receiving allogeneic renal transplants, administered in combination with cyclosporine and corticosteroids. To prevent graft rejection
- MPA also has antibacterial, antifungal, and antiviral activities. is an immunosuppressant drug and **potent anti-proliferative**,

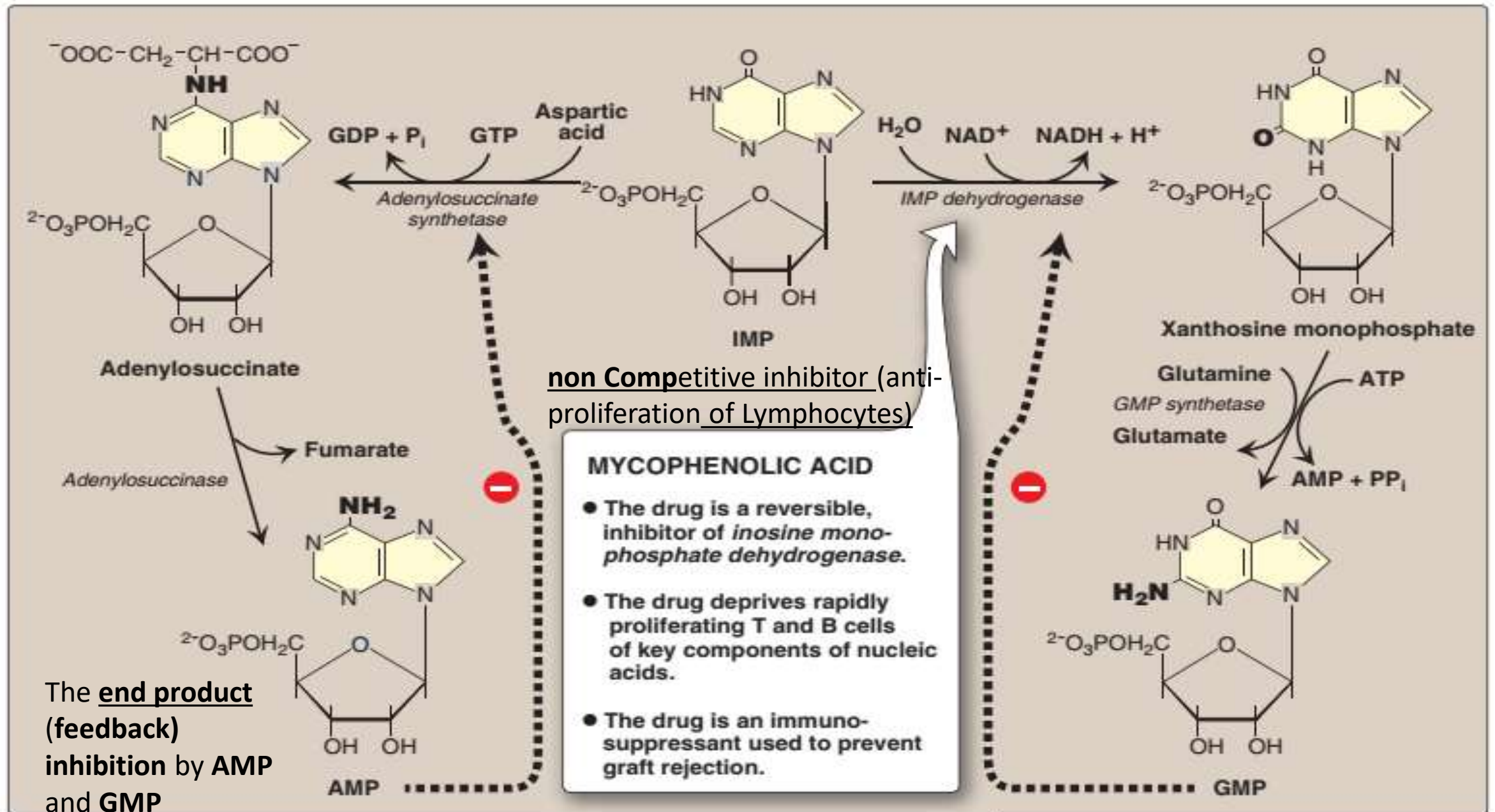


Fig. Conversion of IMP to AMP and GMP showing feedback inhibition

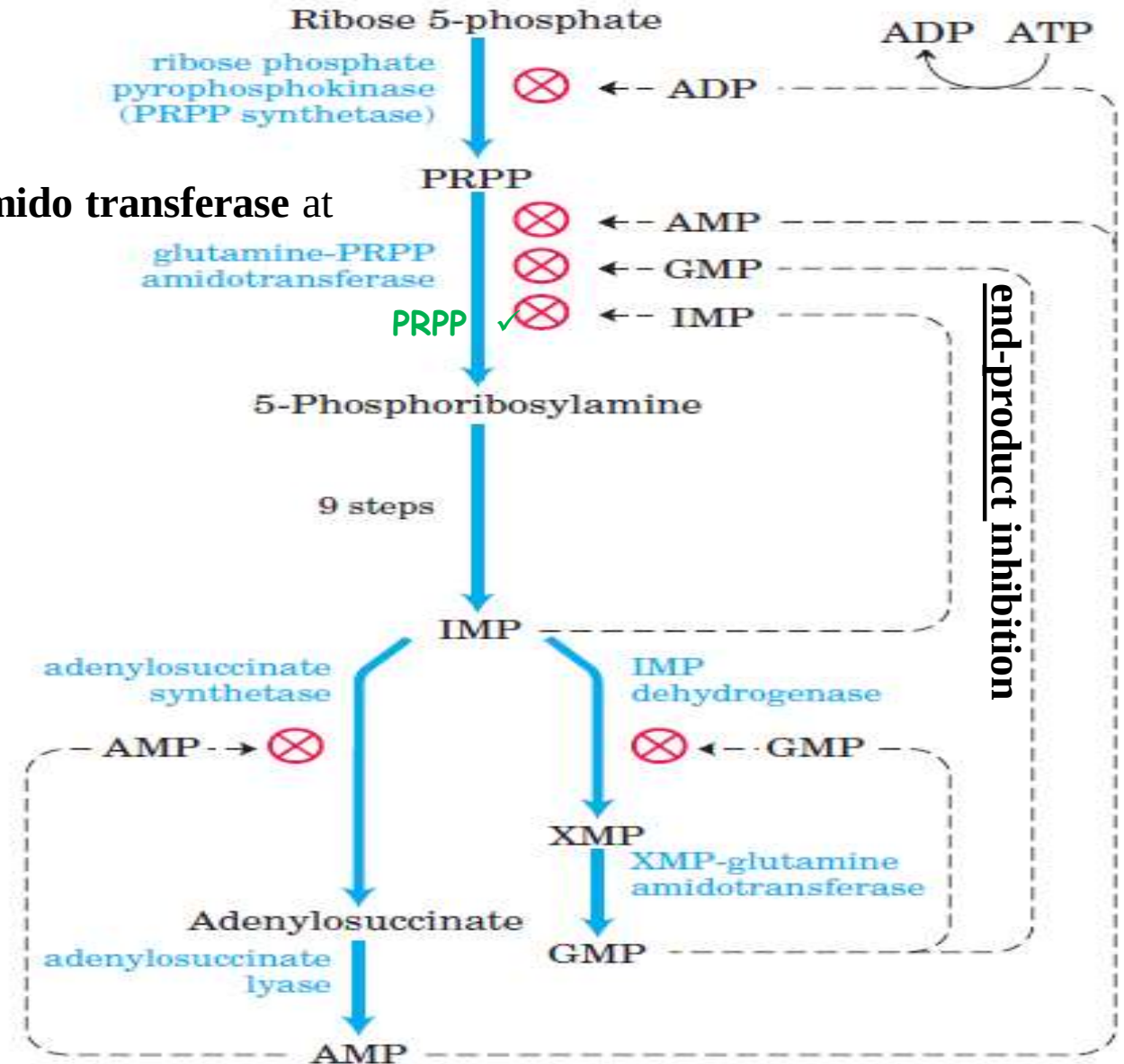
Purine nucleotide biosynthesis is regulated by feedback control

PRPP is a feed-forward activator of ...Amido transferase at step 2

The significance of regulation:

(1) Fulfill the need of the body, without wasting.

(2) $[GTP]=[ATP]$



Nucleoside diphosphate and triphosphate

Nucleoside diphosphate: **ATP-dependent kinase**

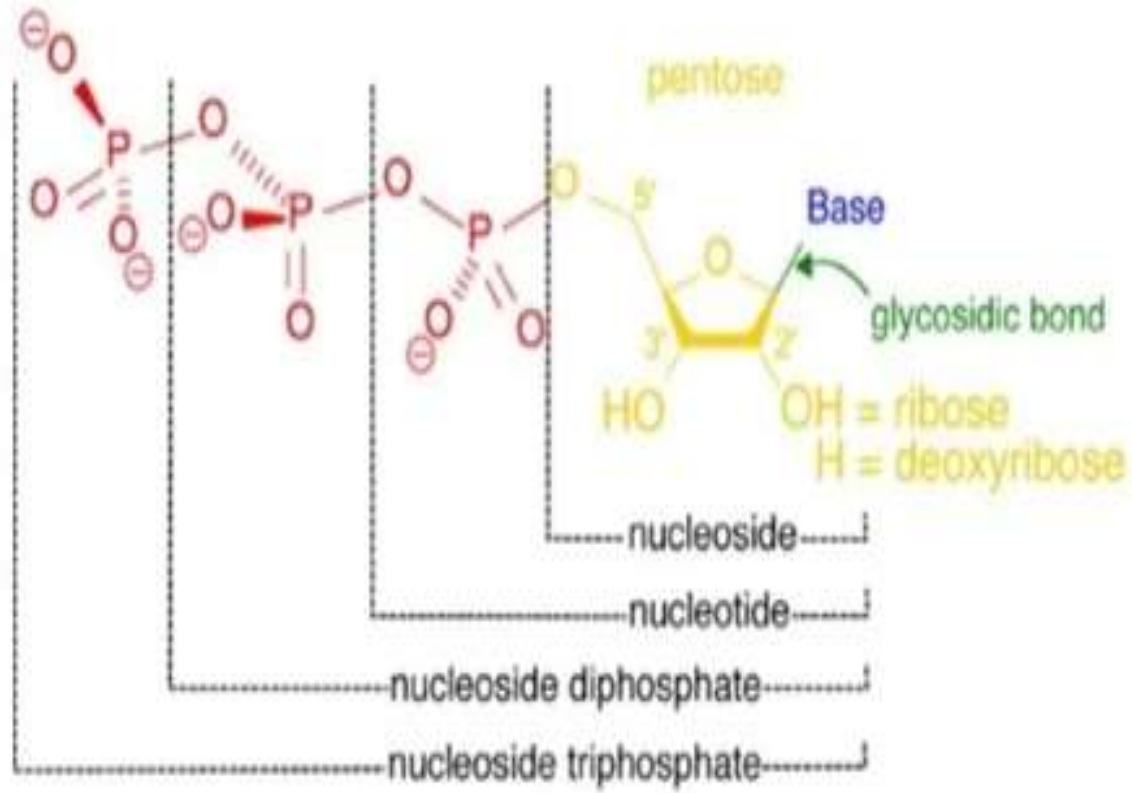
- *Adenylate kinase*: $\text{AMP} + \text{ATP} \rightarrow \text{ADP} + \text{ADP}$
- *Guanylate kinase*: $\text{GMP} + \text{ATP} \rightarrow \text{GDP} + \text{ADP}$

Nucleotide triphosphate: **non-specific enzyme**

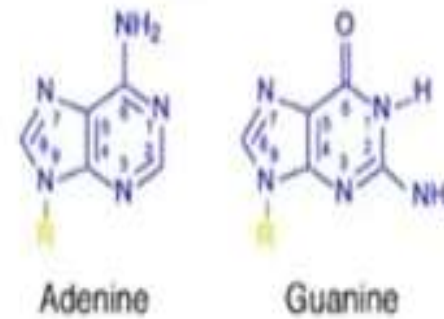
- *Nucleoside diphosphate kinase*



The structure of nucleotides



Purines



How Do Cells Form the Deoxyribonucleotides That Are Necessary for DNA Synthesis?

- Reduction at **2'**-position of ribose ring
- Serve as a precursor for DNA synthesis
- Replacement of 2'-OH with *hydride* is catalyzed by **ribonucleotide reductase**
 - An $\alpha_2\beta_2$ -type enzyme - subunits R_1 (86 kD) and R_2 (43.5 kD)
 - R_1 has two regulatory sites, a **specificity site**, and an overall **activity site**

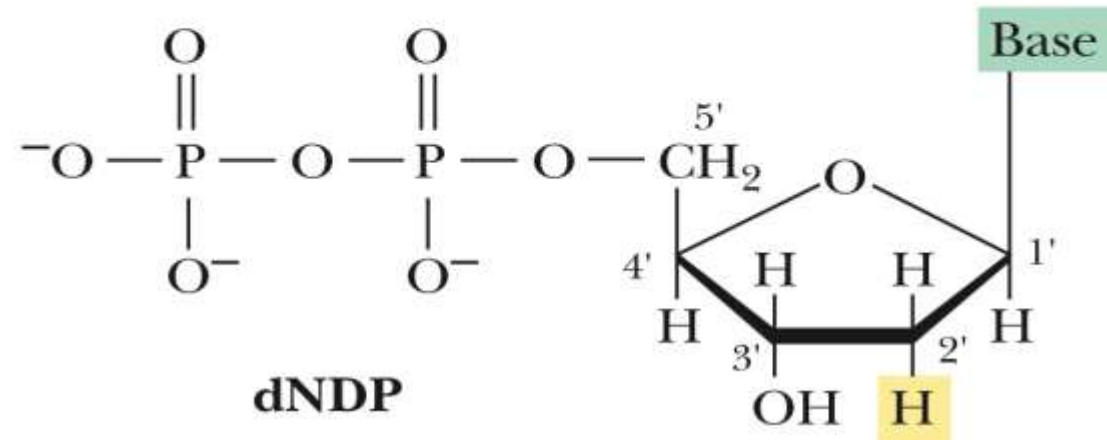
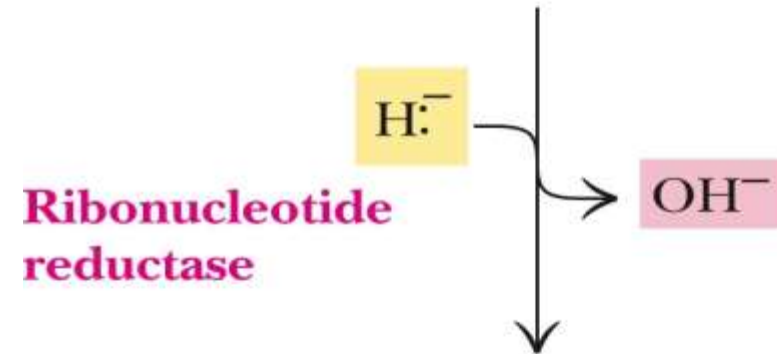
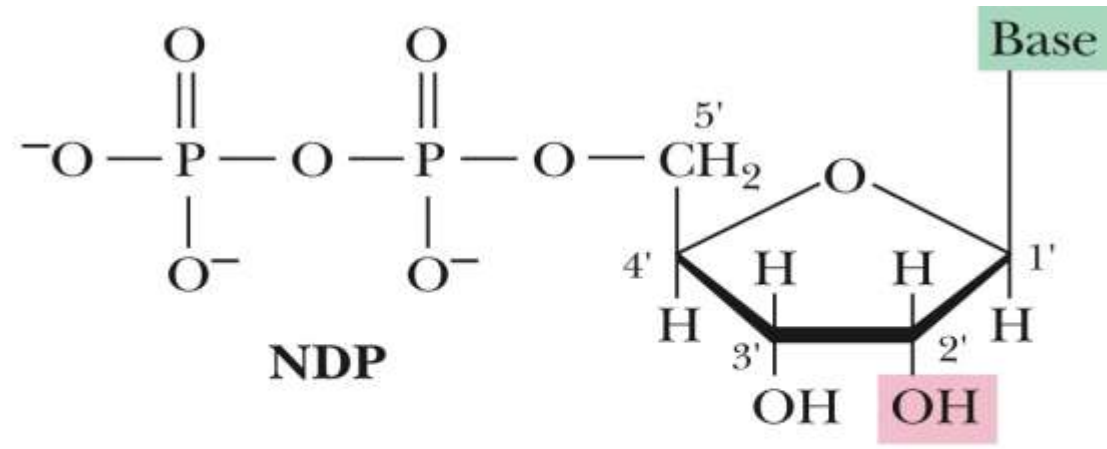


Figure 12
Deoxyribonucleotide synthesis
involves reduction at the 2'-position
of the ribose ring of nucleoside
diphosphates

Regulation of dNTP Synthesis

- The overall activity of **ribonucleotide reductase** must be regulated
 - **ATP activates**, **dATP inhibits** at the overall activity site
- Balance of the four deoxynucleotides must be controlled
 - dCTP, dATP, dTTP, and dGTP bind at the **specificity site**
 - to regulate the selection of **substrates and the products made**

Salvage Pathway of Purine

✓ **Two phosphoribosyl transferases** are involved in this pathway:

- **Adenosine phosphoribosyl transferase (APRT)**

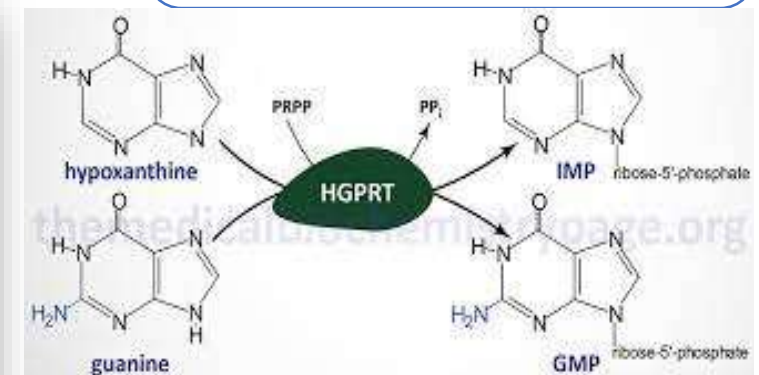
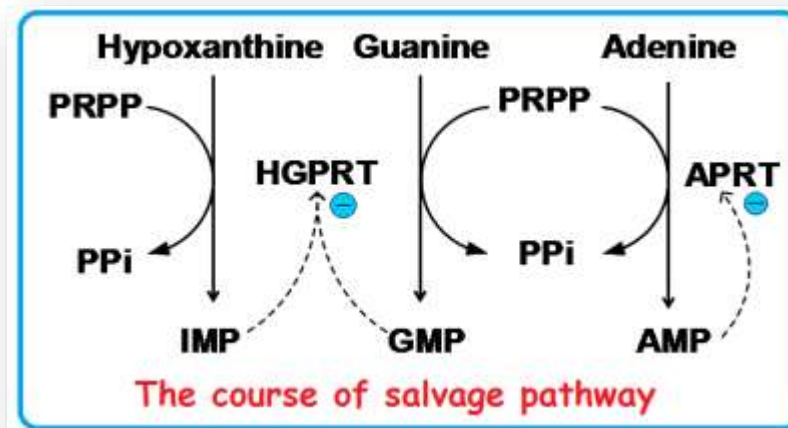


APRT is not very important because it generate little adenine

- **Hypoxanthine-guanine phosphoribosyl transferase (HGPRT)**



HGPRT, is exceptionally important and it is **inhibited by both IMP and GMP**





The significance of salvage pathway :

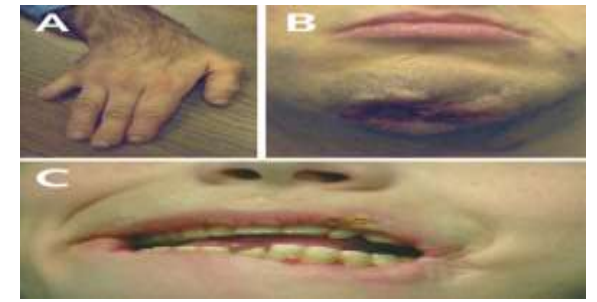
- (1) Save the fuel.
- (2) Some tissues and organs such as brain and bone marrow are only capable of synthesizing nucleotides by a salvage pathway.

Salvage pathway → used to recover bases and nucleosides that are formed during the degradation of RNA and DNA or from an exogenous source (diet) to form nucleotides



Continue...

Biochemistry in medicine



- **Mutations** in **genes** that encode **nucleotide biosynthetic enzymes** can reduce levels of needed nucleotides and can lead to an **accumulation of intermediates**
- **Lesch-Nyhan syndrome** is compulsive **self-destructive behavior**,
 - caused by a **deficiency** of **hypoxanthine-guanine phosphoribosyltransferase (HGPRT)** of salvage pathway,
 - Rare disease is inherited as a **sex-linked recessive disorder**
- In the **absence of HGPRT**,
 - **PRPP levels rise** and **purines** are overproduced by the de novo pathway,
 - resulting in high levels of **uric acid production and gout-like damage to tissue**
- The **brain** is especially **dependent on the salvage pathways**, and
 - this may account for the **central nervous system damage in children with Lesch-Nyhan syndrome**.



Lesch-Nyhan syndrome:

- This syndrome is a rare, X-linked, recessive disorder associated with a virtually complete deficiency of hypoxanthine-guanine phosphoribosyltransferase (HGPRT).
- This deficiency results in an inability to salvage hypoxanthine or guanine, from which excessive amounts of uric acid are produced
- In patients with Lesch-Nyhan syndrome, the hyperuricemia frequently results in the formation of uric acid stones in the kidneys (urolithiasis) and the deposition of urate crystals in the joints (gouty arthritis) and soft tissues.
- In addition, the syndrome is characterized by motor dysfunction, cognitive deficits, and behavioral disturbances that include self-mutilation (biting of lips and fingers)



Figure
Lesions on the lips of Lesch-Nyhan patients caused by self-mutilation.



urine

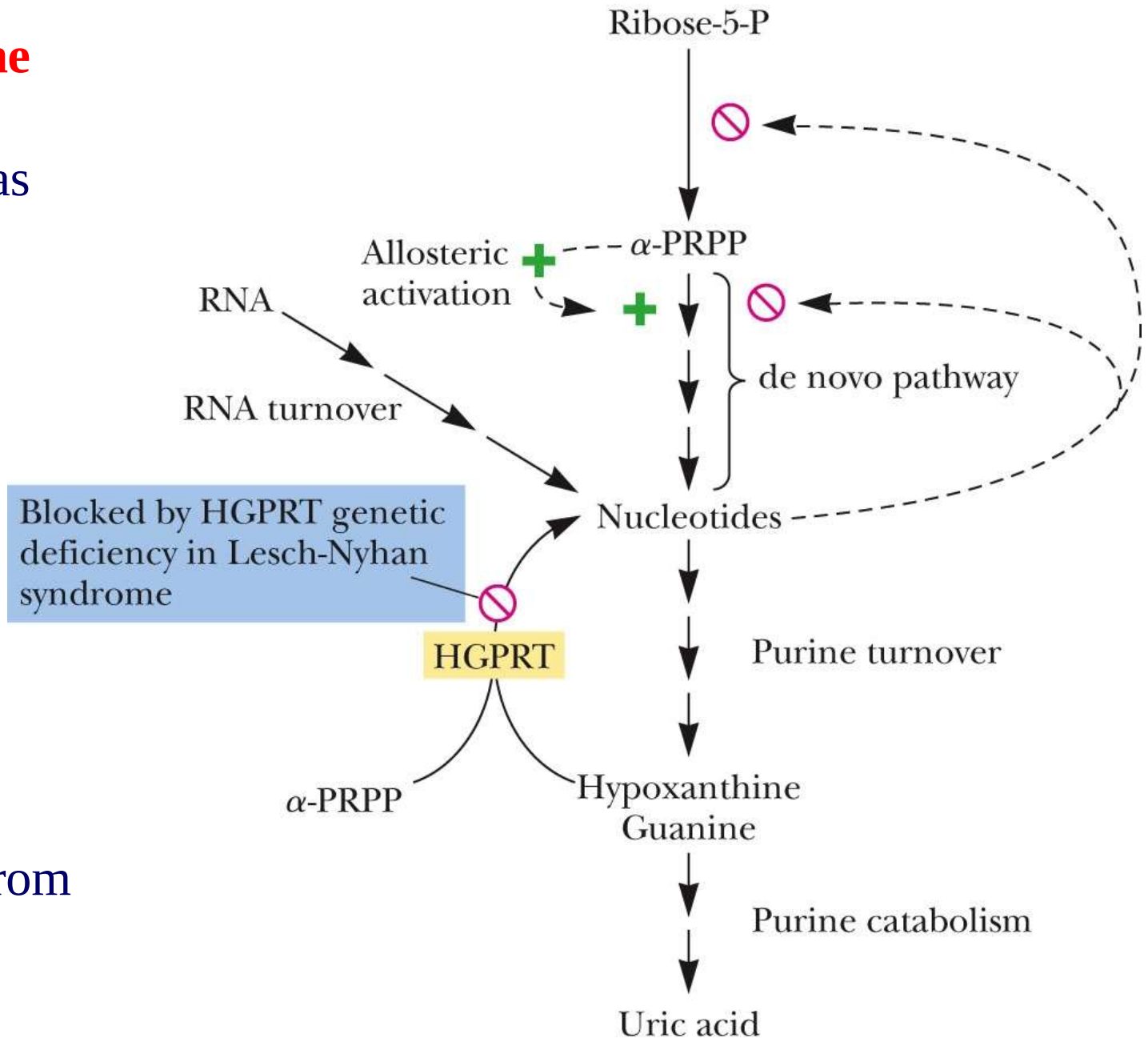
Victims of **Lesch-Nyhan syndrome** experience **severe arthritis** due to accumulation of uric acid, as well as retardation, and other neurological symptoms.

Cause → (HGPRT) Gene **MUTATED**

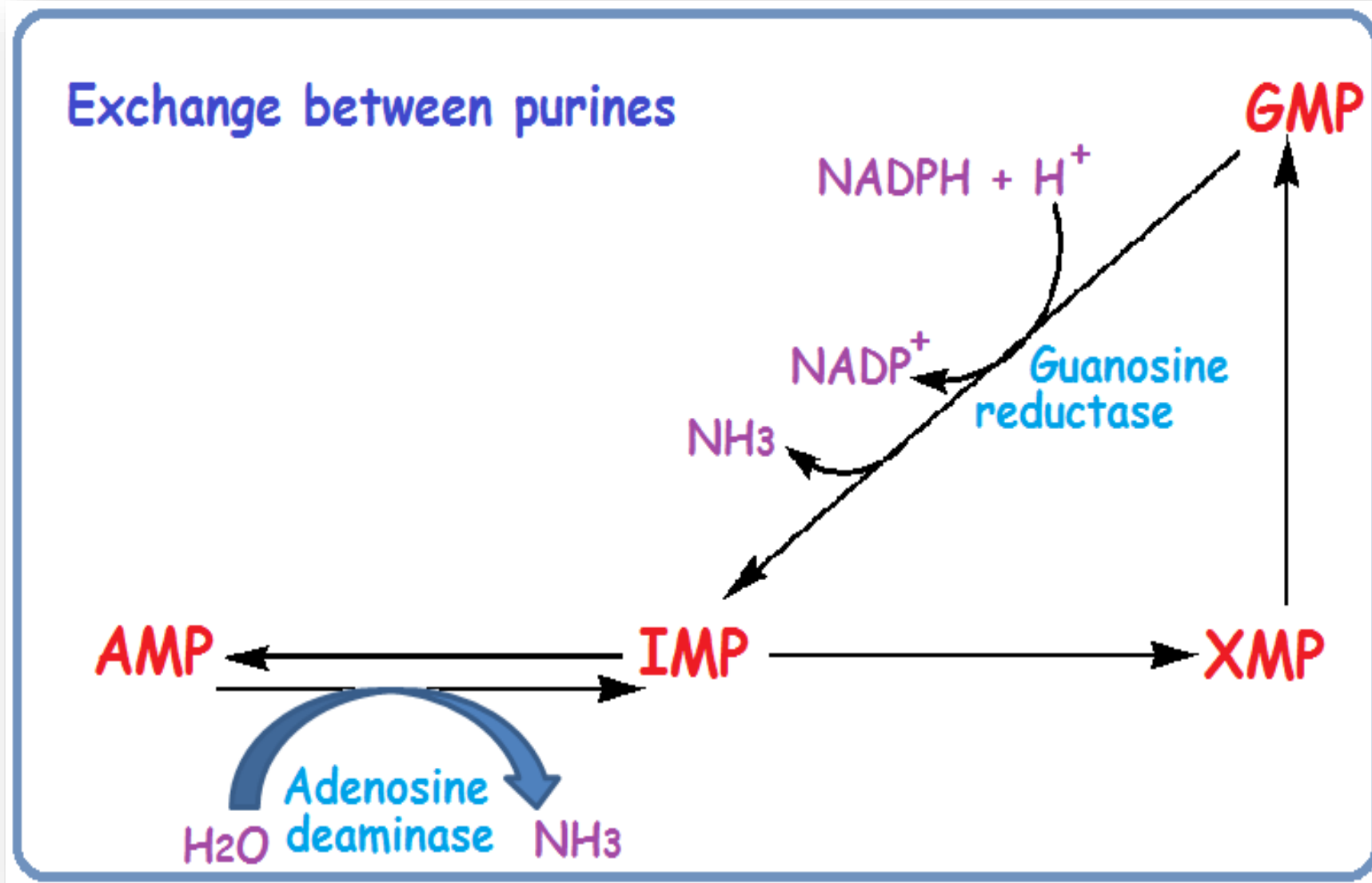
Pathophysiology → IMP and GMP level decrease and increased PRPP level cause stimulation of **De novo- purine synthesis** pathway

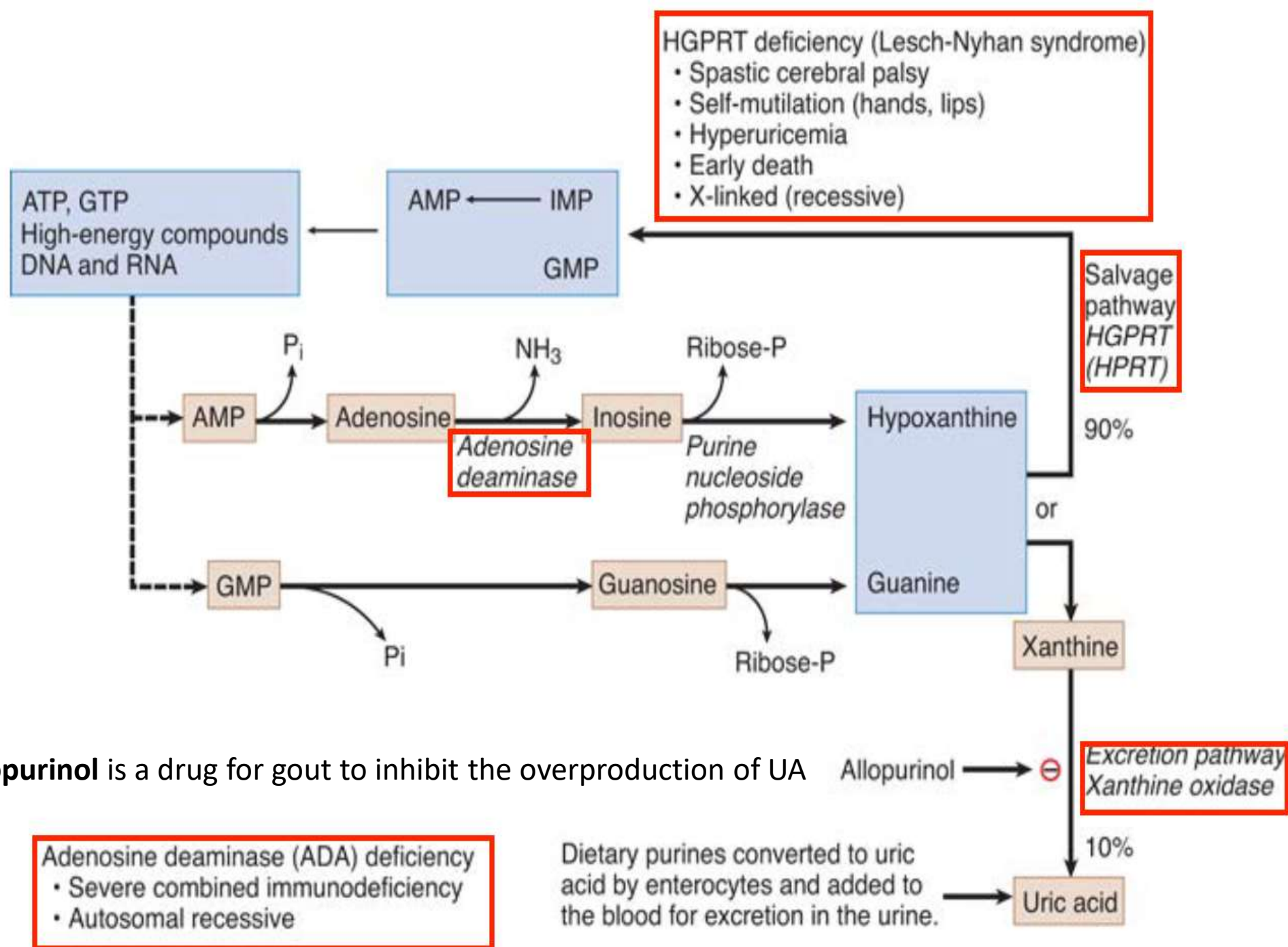
“PRPP High, first precursor (**feed-forward activator**) for **De novo**, **feedback inhibitor** of de novo (IMP and GMP) reduced”

Lesch-Nyhan syndrome results from a **complete deficiency in HGPRT**



Continue...





Allopurinol is a drug for gout to inhibit the overproduction of UA

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Thank You!!